Lab 01: Software Requirement Specification

1. Scope

The scope of this project is to design and develop a web-based Online Fashion Store that enables customers to browse, select, and purchase fashion products online, while allowing the admin to manage the entire platform.

On the customer side, the system will allow users to register, log in, browse products by category, add items to a cart, place orders, make payments through multiple options, track their order status, and request returns/refunds if needed.

On the admin side, the system will allow management of products, categories, inventory, customer orders, payments, and returns, along with generating basic sales reports.

The system is limited to a single-vendor fashion store (i.e., one business selling its own products) and does not include multi-vendor/marketplace functionality, third-party courier integration, or a dedicated delivery rider module — these are considered out of scope and may be added in future versions.

2. Functional Requirements

- Customers can register and log in to their account.
- Customers can browse products by category (men, women, kids, etc.).
- Customers can search and filter products (by size, color, price).
- Customers can view product details (images, price, description, sizes).
- Customers can add/remove products to/from the shopping cart.
- Customers can place an order and choose a payment method.
- Customers can make payment via multiple options (card, COD, wallet).
- Customers can track the status of their order (pending, shipped, delivered).
- Customers can view order history and download invoices/receipts.
- Customers can request a return/refund for a delivered item.
- Customers can edit their profile and shipping address.
- Customers can leave reviews and ratings on purchased products.
- Admin can add, edit, or remove products and categories.
- Admin can manage inventory and stock levels.
- Admin can view and manage customer orders (confirm, ship, cancel).

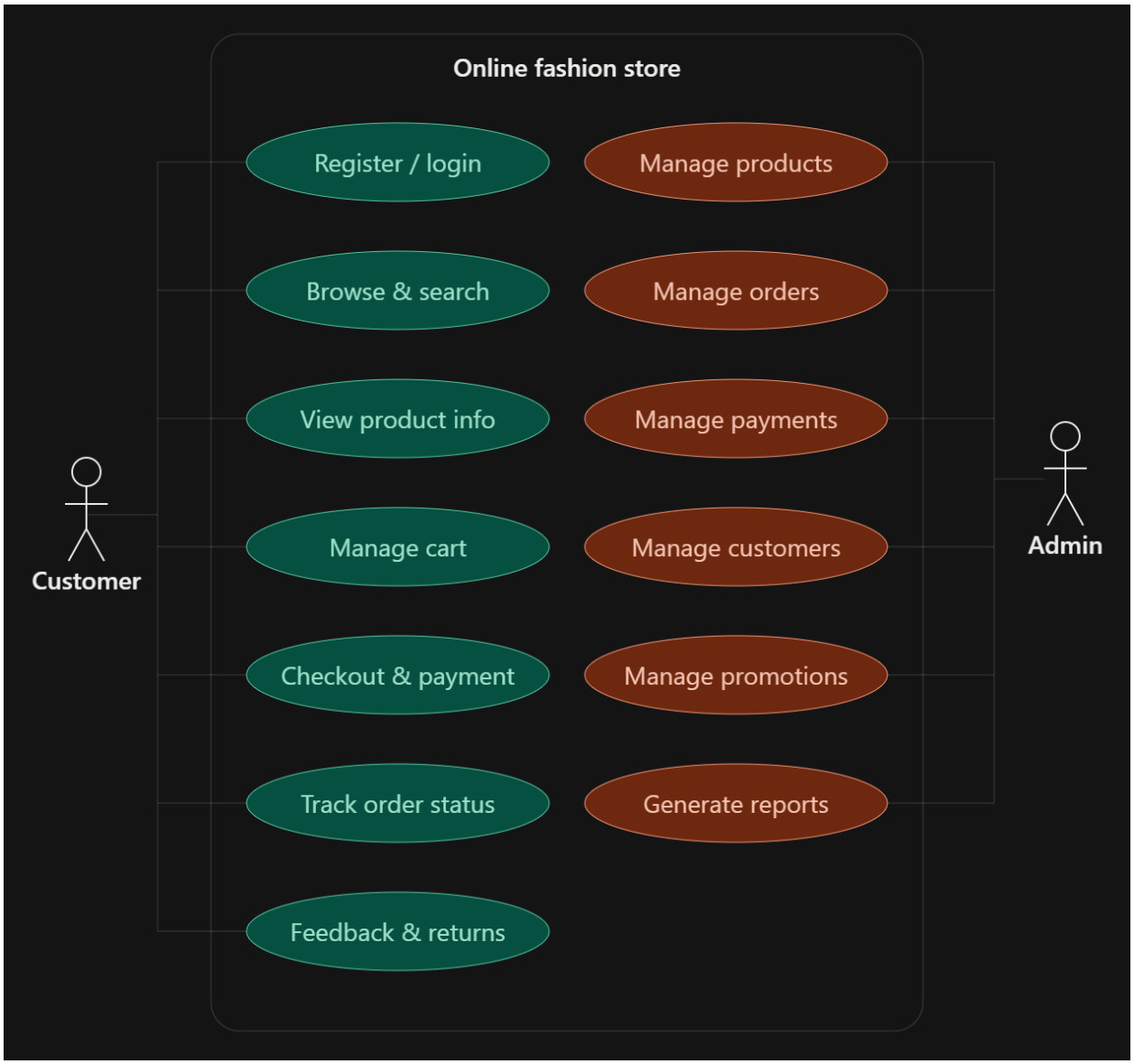
- Admin can update order status for tracking purposes.
- Admin can manage payments and process refunds.
- Admin can manage customer accounts.
- Admin can generate sales and inventory reports.
- Admin can manage discounts/promotional offers.

3. Non-Functional Requirements

- The system will load pages within 2–3 seconds.
- The system will support multiple users browsing/ordering at the same time.
- Payment and customer data must be securely encrypted.
- Only authorized admins should have access to the admin panel.
- The system will be available 24/7 for customers to shop anytime.
- The interface will be simple and user-friendly for all age groups.
- The system will work properly on mobile, tablet, and desktop devices.
- The system will handle errors gracefully without losing order/payment data.
- The system will be scalable to support growing products and users in future.
- The codebase will be maintainable for future updates.
- Order and payment records must remain accurate and consistent at all times.

Lab 2:

Use case diagram



Customer Use Cases

UC1: Register

UC I'd	01
UC Name	Register
Actor	Customer
Description	New customer creates an account on the system
Precondition	Customer has internet access and is on the signup page
Postcondition	New customer account is created
Action	1. Customer selects "Register" 2. Customer enters name, email, and password. 3. System validates the entered information. 4. System creates the new account. 5. System redirects customer to homepage/login.
Alternative Flow	Customer registers using a social media account (Google/Facebook) instead of email.
Exception	If the email is already registered or fields are invalid/empty, system displays an error message and does not create the account.

UC2: Login

UC I'd	02
UC Name	Login
Actor	Customer
Description	Existing customer logs into their account

Precondition	Customer has a registered account and is on the login page
Postcondition	Customer is authenticated and redirected to homepage
Action	1. Customer enters email and password. 2. Customer clicks "Login". 3. System validates credentials. 4. System grants access and redirects to homepage.
Alternative Flow	Customer selects "Forgot Password" to reset credentials.
Exception	If email/password is incorrect, system displays "Invalid credentials" and denies access.

UC3: Browse & Search

Uc I'd	03
UC Name	Browse & Search
Actor	Customer
Description	Customer browses products by category or searches/filters them
Precondition	Customer is on the store homepage
Postcondition	Matching products are displayed
Action	1. Customer selects a category or enters a search keyword. 2. Customer applies filters (size, color, price). 3. System fetches matching products. 4. System displays product list.
Alternative Flow	Customer clears filters to reset the search.
Exception	If no products match, system displays "No results found".

UC4: View Product Info

UC I’d	04
UC Name	View Product Info
Actor	Customer
Description	Customer views detailed information about a specific product
Precondition	Customer has browsed/searched and selected a product
Postcondition	Product detail page is displayed
Action	1. Customer clicks on a product. 2. System retrieves product details (images, price, sizes, description, reviews). 3. System displays the product detail page.
Alternative Flow	Customer scrolls through product images or reads reviews.
Exception	If the product is out of stock, system displays "Currently unavailable" and disables "Add to Cart".

UC5: Manage Cart

UC I’d	05
UC Name	Manage Cart
Actor	Customer
Description	Customer adds, updates, or removes products from the shopping cart
Precondition	Customer is logged in and viewing a product
Postcondition	Cart is updated with selected items
Action	1. Customer selects size/quantity. 2. Customer clicks "Add to Cart". 3. System updates cart and shows confirmation. 4. Customer opens cart and edits quantity or removes items.

Alternative Flow	Customer saves an item to wishlist instead of cart.
Exception	If item goes out of stock before checkout, system notifies customer and removes it from cart.

UC6: Checkout & Payment

UC I'd	06
UC Name	Checkout & Payment
Actor	Customer
Description	Customer places an order and makes payment for items in cart
Precondition	Customer is logged in and has items in the cart
Postcondition	Order is placed and payment is confirmed
Action	1. Customer clicks "Checkout". 2. System displays order summary (items, total price). 3. Customer enters/confirms shipping address. 4. Customer selects a payment method. 5. Customer confirms payment. 6. System processes payment and generates order confirmation.
Alternative Flow	Customer applies a promo/discount code before confirming payment.
Exception	If payment fails, system displays an error and allows customer to retry or choose another payment method.

UC7: Track Order Status

UC I'd	07
UC Name	Track Order Status
Actor	Customer

Description	Customer checks the current status of a placed order
Precondition	Customer has placed at least one order
Postcondition	Customer views updated order status
Action	1. Customer logs in and goes to "My Orders". 2. System displays list of orders. 3. Customer selects an order. 4. System shows current status (Pending/Shipped/Delivered).
Alternative Flow	Customer downloads the invoice/receipt for the selected order.
Exception	If order not found, system displays "No such order exists".

UC8: Feedback & Returns

UC I'd	08
UC Name	Feedback & Returns
Actor	Customer
Description	Customer submits feedback/rating or requests a return/refund for a delivered item
Precondition	Customer has a delivered order
Postcondition	Feedback is recorded or return request is submitted
Action	1. Customer goes to order history. 2. Customer selects a delivered item. 3. Customer chooses "Give Feedback" or "Request Return". 4. For feedback: enters rating and comments. 5. For return: selects reason and submits request. 6. System saves feedback/forwards return request to admin.
Alternative Flow	Customer attaches photos to support the return request.
Exception	If return window has expired, system rejects the return request.

Admin Use Cases

UC9: Manage Products

UC I'd	09
UC Name	Manage Products
Actor	Admin
Description	Admin adds, edits, or removes products from the store
Precondition	Admin is logged in
Postcondition	Product catalog is updated
Action	1. Admin opens "Products" panel. 2. Admin selects Add/Edit/Delete. 3. Admin enters product details (name, price, size, image). 4. System validates and saves the product. 5. Product becomes visible to customers.
Alternative Flow	Admin bulk-uploads products via a file.
Exception	If required fields are missing or invalid, system shows a validation error and does not save.

UC10: Manage Orders

UC I'd	10
UC Name	Manage Orders
Actor	Admin
Description	Admin views and updates the status of customer orders
Precondition	Admin is logged in

Postcondition	Order status is updated in the system
Action	1. Admin opens "Orders" panel. 2. System displays all customer orders. 3. Admin selects an order. 4. Admin updates status (confirm/ship/deliver/cancel). 5. System saves the updated status and notifies customer.
Alternative Flow	Admin filters orders by status or date before selecting one.
Exception	If admin cancels an order with completed payment, system automatically triggers a refund process.

UC11: Manage Payments

UC I'd	11
UC Name	Manage Payments
Actor	Admin
Description	Admin views payment records and processes refunds
Precondition	Admin is logged in
Postcondition	Payment status is updated or refund is processed
Action	1. Admin opens "Payments" panel. 2. System displays list of transactions. 3. Admin selects a transaction (e.g., for a return/cancellation). 4. Admin initiates refund. 5. System processes refund and updates payment status.
Alternative Flow	Admin exports the payment records for accounting.
Exception	If refund processing fails, system logs the error and notifies admin to retry.

UC12: Manage Customers

UC I’d	12
UC Name	Manage Customers
Actor	Admin
Description	Admin views and manages customer accounts
Precondition	Admin is logged in
Postcondition	Customer account is updated, suspended, or removed
Action	1. Admin opens "Customers" panel. 2. System displays list of registered customers. 3. Admin selects a customer. 4. Admin views profile/order history, or suspends/deletes account.
Alternative Flow	Admin resets a customer's password on request.
Exception	If admin tries to delete a customer with active orders, system blocks deletion and shows a warning.

UC13: Manage Promotions

UC I’d	13
UC Name	Manage Promotions
Actor	Admin
Description	Admin creates and manages discount codes/promotional offers
Precondition	Admin is logged in
Postcondition	New promotion is active or existing one is updated
Action	1. Admin opens "Promotions" panel. 2. Admin creates a new discount (code, percentage, validity dates). 3. System validates and activates the promotion. 4. Promotion becomes visible/applicable to customers at checkout.

Alternative Flow	Admin deactivates or extends an existing promotion.
Exception	If discount code already exists, system shows a duplicate error and does not save.

UC14: Generate Reports

UC I'd	14
UC Name	Generate Reports
Actor	Admin
Description	Admin generates sales and inventory reports for business insights
Precondition	Admin is logged in and sufficient order/sales data exists
Postcondition	Report is generated and displayed/downloaded
Action	1. Admin opens "Reports" panel. 2. Admin selects report type (sales, inventory, top products) and date range. 3. System compiles data. 4. System displays/downloads the report.
Alternative Flow	Admin exports the report as PDF/Excel.
Exception	If no data exists for the selected range, system displays "No data available".

SYSTEM DESIGN & PRESENTATION SPECIFICATION

ONLINE FASHION STORE

UML Sequence Diagrams & Use Case Interaction Manual

A complete, presentation-ready UML Sequence Diagram specification for the Online Fashion Store system. Each diagram maps directly to the project's Use Case specifications, illustrating the dynamic interactions between actors (Customer / Admin), User Interface (UI), System Controller, and Database. Specifically structured for academic project documentation, slide presentations, and viva defense.

Project Title	Online Fashion Store (Ecommerce-Fashion)
System Domain	Single-Vendor Web-Based E-Commerce Platform
Diagram Modeling Level	Academic Presentation Level (Actor: User -> Screen: GUI -> FashionDB: MSSQL)
Document Version	2.0 (Presentation Edition)
Requirements Coverage	100% Coverage of Lab 01 Functional Requirements (FR-01 to FR-20)
Documentation Scope	All Customer & Admin Use Cases (UC-01 through UC-15)

1. Sequence Diagram Presentation Pattern

In software project presentations, final viva defenses, and academic evaluation panels, sequence diagrams must clearly communicate the chronological flow of actions for each use case. Rather than cluttering diagrams with low-level implementation details or distributed

network frames, the presentation-level pattern models the essential interaction flow between the user, the graphical user interface, and the database.

1.1 Academic Presentation Architecture (Actor -> GUI -> Database)

Lifeline Element	UML Role	Presentation & Viva Explanation
Actor: User (Customer / Admin)	Human Initiator	The actor initiating the use case via client web browser or administrative management console.
<Module>: GUI (Boundary)	Presentation Boundary	The specific user interface screen or modal form (e.g., Register: GUI, Cart: GUI, ProductManager: GUI) capturing input and displaying responses.
FashionDB: MSSQL (Data Store)	Persistent Entity Store	The relational database management system maintaining products, customer accounts, orders, and configuration records.
PaymentGateway: System (External)	Financial Service	External banking/payment gateway service processing electronic card charges and automated refunds (UC-06 & UC-11).

2. Use Case Sequence Diagrams (Complete Catalog)

The following catalog details all Use Cases for the Online Fashion Store. Each section contains the use case description, actor, preconditions, postconditions, an embedded presentation sequence diagram visual, a step-by-step trace table, and alternative and exception branch analysis.

2.1 UC-01: Register

Description: Customer registers for a new store account by opening the registration form, providing personal information (name, email, password, contact), validating email uniqueness against the database, and saving the customer profile.

Actor	Customer
Trigger	Customer navigates to the registration screen and submits their registration details.
Preconditions	• Customer has accessed the online fashion store web portal. • Customer has a valid, unique email address.

Postconditions	• New customer record is saved in FashionDB: MSSQL. • Customer is registered and redirected to login/homepage.

Figure 2.1: Sequence Diagram — UC-01: Register

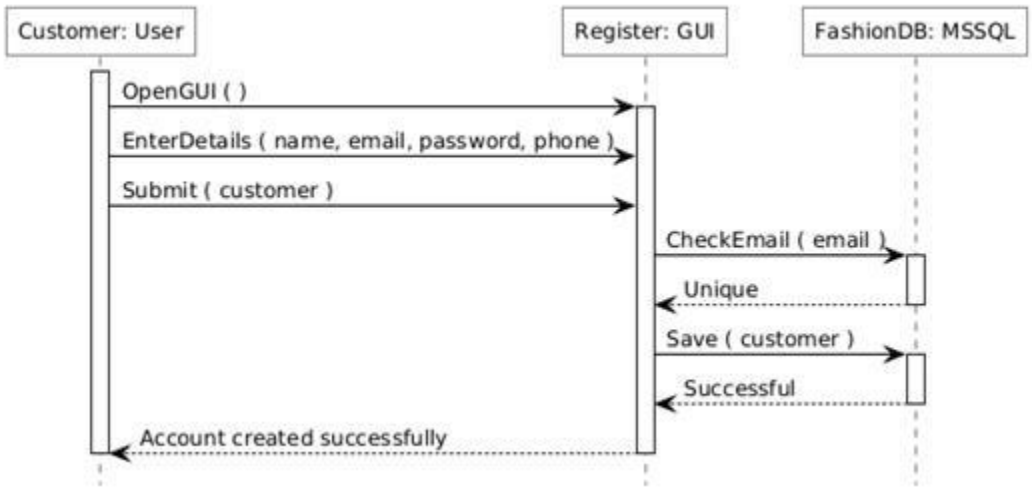


Table 2.1: Action Steps & Presentation Flow for UC-01

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Customer: User	Register: GUI	**OpenGUI ( )**	Normal	Customer clicks register link on store navigation header.
2	Customer: User	Register: GUI	**EnterDetails (name, email, password, phone)**	Normal	Customer fills out the account registration input fields.
3	Customer: User	Register: GUI	**Submit (customer)**	Normal	Customer clicks the submit button to register.
4	Register: GUI	FashionDB: MSSQL	**CheckEmail (email)**	Normal	GUI queries the database to verify email uniqueness.
5	FashionDB: MSSQL	Register: GUI	**Unique**	Normal	Database confirms the provided email is not previously registered.
6	Register: GUI	FashionDB: MSSQL	**Save (customer)**	Normal	GUI transmits customer entity to be inserted into database.

7	FashionDB: MSSQL	Register: GUI	**Successful**	Normal	Database returns success status for customer insertion.
8	Register: GUI	Customer: User	**Account created successfully**	Normal	GUI displays confirmation message and redirects customer.

Alternative Branches & Exceptions:

- Social Media Registration: Customer selects 'Sign up with Google'. GUI authenticates external identity and saves profile directly in FashionDB: MSSQL.
- Duplicate Email: If email already exists, FashionDB returns 'Email Exists', and Register: GUI displays 'Email already registered. Please login'.

2.2 UC-02: Login

Description: Customer logs in to their existing account by providing registered email and password credentials, which are verified against the database before granting account access.

Actor	Customer
Trigger	Customer enters credentials and clicks 'Login'.
Preconditions	• Customer possesses an existing registered account in FashionDB: MSSQL. • Customer has accessed the login screen.
Postconditions	• Customer session is authenticated and authorized. • Customer receives access to account features and personalized dashboard.

Figure 2.2: Sequence Diagram — UC-02: Login

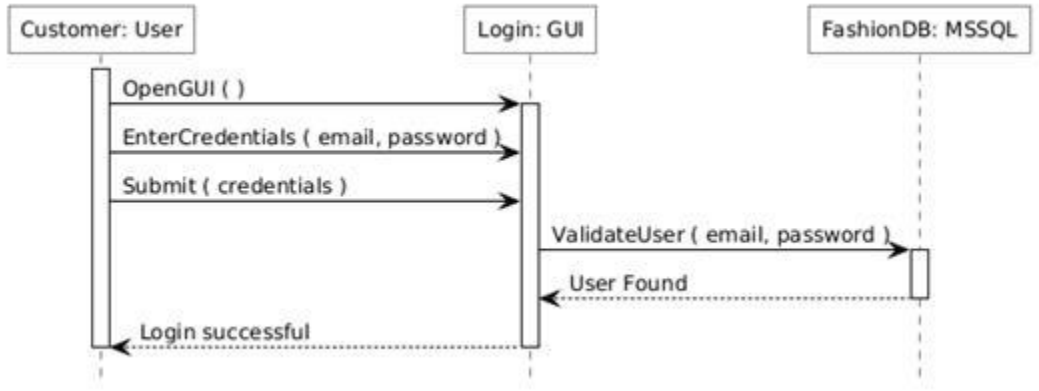


Table 2.2: Action Steps & Presentation Flow for UC-02

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation

1	Customer: User	Login: GUI	**OpenGUI ( )**	Normal	Customer clicks login button on store navigation bar.
2	Customer: User	Login: GUI	**EnterCredentials (email, password)**	Normal	Customer types email and password into login fields.
3	Customer: User	Login: GUI	**Submit (credentials)**	Normal	Customer clicks the login submit button.
4	Login: GUI	FashionDB: MSSQL	**ValidateUser (email, password)**	Normal	GUI queries database to verify user credentials.
5	FashionDB: MSSQL	Login: GUI	**User Found**	Normal	Database verifies credentials and returns matching user profile.
6	Login: GUI	Customer: User	**Login successful**	Normal	GUI creates user session and redirects customer to homepage.

Alternative Branches & Exceptions:

- Password Reset Flow: Customer clicks 'Forgot Password'. Login GUI prompts for email and triggers verification link.
- Invalid Credentials: If email or password is invalid, FashionDB returns 'Invalid', and Login: GUI displays 'Invalid credentials. Please try again'.

2.3 UC-03: Browse and Search

Description: Customer explores fashion products by selecting categories (men, women, kids) and entering search queries with faceted filters (price, size, color) to retrieve matching products from the database.

Actor	Customer
Trigger	Customer selects a catalog category or inputs search keywords.
Preconditions	• Fashion store catalog is populated with active apparel items in FashionDB: MSSQL.
Postconditions	• Filtered fashion product catalog is presented on the customer screen.

Figure 2.3: Sequence Diagram — UC-03: Browse and Search

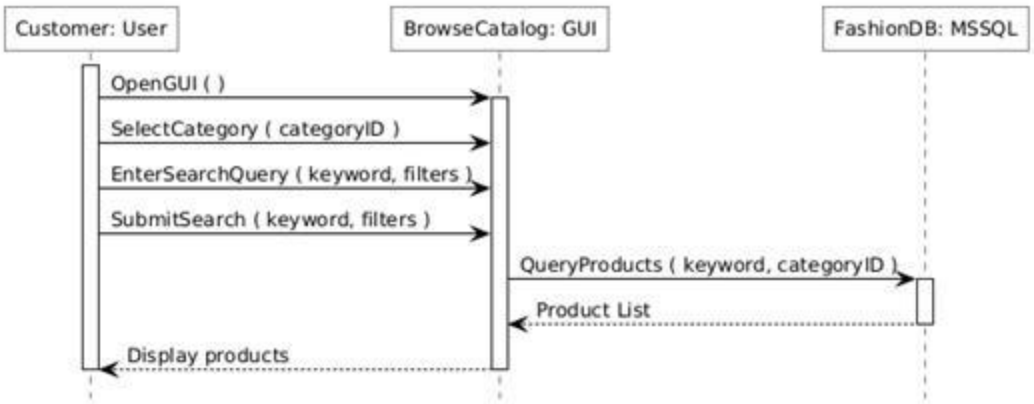


Table 2.3: Action Steps & Presentation Flow for UC-03

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Customer: User	BrowseCatalog: GUI	**OpenGUI ( )**	Normal	Customer accesses the store storefront / catalog page.
2	Customer: User	BrowseCatalog: GUI	**SelectCategory (categoryID)**	Normal	Customer selects apparel category (e.g., Men, Women, Kids).
3	Customer: User	BrowseCatalog: GUI	**EnterSearchQuery (keyword, filters)**	Normal	Customer specifies search term, size, color, or price range.
4	Customer: User	BrowseCatalog: GUI	**SubmitSearch (keyword, filters)**	Normal	Customer triggers the search filtering operation.
5	BrowseCatalog: GUI	FashionDB: MSSQL	**QueryProducts (keyword, categoryID)**	Normal	GUI queries product catalog table in FashionDB.
6	FashionDB: MSSQL	BrowseCatalog: GUI	**Product List**	Normal	Database returns list of matched product records.
7	BrowseCatalog: GUI	Customer: User	**Display products**	Normal	GUI displays product grid matching customer criteria.

Alternative Branches & Exceptions:

- Sorting by Price/Popularity: Customer selects sort criteria (e.g. Price: Low to High). GUI re-queries database with sorting parameters.
- No Products Found: If query matches zero products, FashionDB returns empty set, and GUI displays 'No products found matching your filters'.

2.4 UC-04: View Product Info

Description: Customer clicks on a specific product card to view comprehensive item specifications, high-definition image galleries, available sizes, stock status, pricing, and authentic customer reviews retrieved from the database.

Actor	Customer
Trigger	Customer clicks on a product card or thumbnail.
Preconditions	• Customer is browsing the catalog or search results. • Selected product exists in FashionDB: MSSQL.
Postconditions	• Detailed product view and customer ratings/reviews are displayed.

Figure 2.4: Sequence Diagram — UC-04: View Product Info

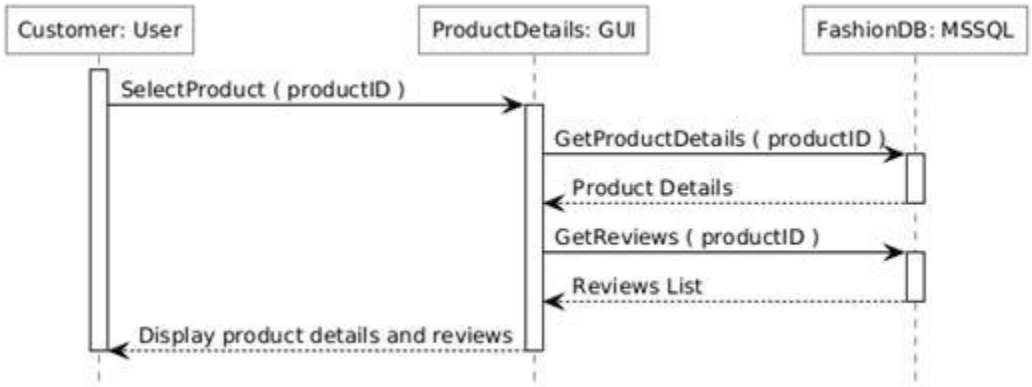


Table 2.4: Action Steps & Presentation Flow for UC-04

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Customer: User	ProductDetails: GUI	**SelectProduct (productID)**	Normal	Customer clicks on an apparel item card in the catalog.
2	ProductDetails: GUI	FashionDB: MSSQL	**GetProductDetails (productID)**	Normal	GUI fetches detailed product attributes from database.
3	FashionDB: MSSQL	ProductDetails: GUI	**Product Details**	Normal	Database returns images, description, sizes, and price.

4	ProductDetails: GUI	FashionDB: MSSQL	**GetReviews (productID)**	Normal	GUI queries ratings and customer feedback for the product.
5	FashionDB: MSSQL	ProductDetails: GUI	**Reviews List**	Normal	Database returns list of approved customer reviews.
6	ProductDetails: GUI	Customer: User	**Display product details and reviews**	Normal	GUI presents detailed product view with image gallery and reviews.

Alternative Branches & Exceptions:

- Zoom & Gallery View: Customer clicks to inspect high-resolution fabric texture images.
- Product Inactive/Removed: If item is discontinued, FashionDB returns null, and GUI displays 'This product is no longer available'.

2.5 UC-05: Manage Cart

Description: Customer adds fashion items to their shopping cart, verifies item inventory availability in the database, adjusts item quantities, or removes items prior to proceeding to checkout.

Actor	Customer
Trigger	Customer clicks 'Add to Cart', adjusts item quantity, or opens the cart view.
Preconditions	• Customer is viewing a product page or cart drawer. • Items added have valid size and quantity selections.
Postconditions	• Shopping cart items and running total are updated and saved in FashionDB: MSSQL.

Figure 2.5: Sequence Diagram — UC-05: Manage Cart

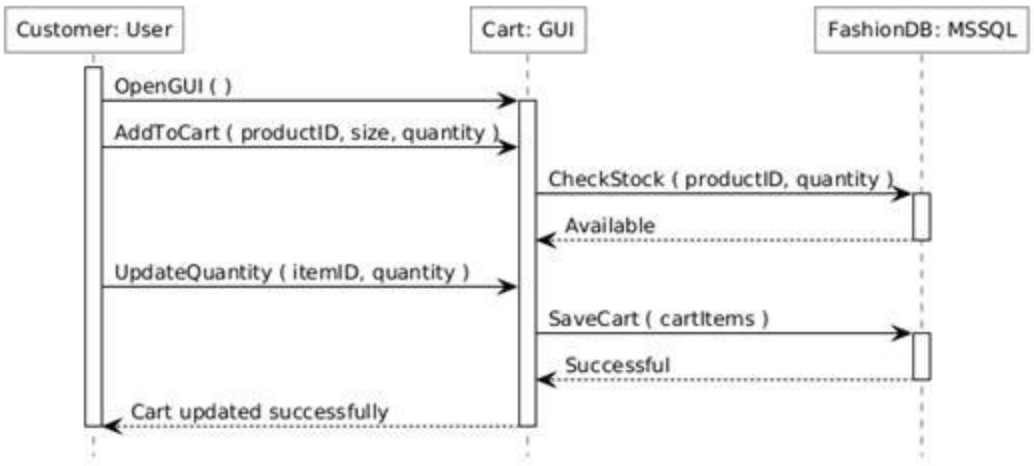


Table 2.5: Action Steps & Presentation Flow for UC-05

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Customer: User	Cart: GUI	**OpenGUI ( )**	Normal	Customer opens the shopping cart panel.
2	Customer: User	Cart: GUI	**AddToCart (productID, size, quantity)**	Normal	Customer adds apparel item with chosen size and quantity.
3	Cart: GUI	FashionDB: MSSQL	**CheckStock (productID, quantity)**	Normal	GUI queries stock level in database.
4	FashionDB: MSSQL	Cart: GUI	**Available**	Normal	Database confirms required inventory is in stock.
5	Customer: User	Cart: GUI	**UpdateQuantity (itemID, quantity)**	Normal	Customer adjusts item count or removes item.
6	Cart: GUI	FashionDB: MSSQL	**SaveCart (cartItems)**	Normal	GUI persists updated cart contents in database.
7	FashionDB: MSSQL	Cart: GUI	**Successful**	Normal	Database confirms cart changes are saved.
8	Cart: GUI	Customer: User	**Cart updated successfully**	Normal	GUI updates subtotal and displays updated cart list.

Alternative Branches & Exceptions:

- Remove Item from Cart: Customer clicks 'Remove'. GUI updates cart session and refreshes subtotal.
- Out of Stock: If requested quantity exceeds warehouse stock, FashionDB returns 'Insufficient Stock', and Cart: GUI displays warning.

2.6 UC-06: Checkout and Payment

Description: Customer provides shipping address, selects payment method (Credit/Debit Card, Cash on Delivery, Mobile Wallet), authorizes transaction through the payment gateway, and persists the confirmed order in the database.

Actor	Customer
Trigger	Customer clicks 'Proceed to Checkout' and submits payment details.
Preconditions	• Customer has one or more valid items in their shopping cart. • Customer provides complete delivery and payment information.
Postconditions	• Order is confirmed with unique Order ID and payment transaction record in FashionDB: MSSQL. • Inventory stock is decremented accordingly.

Figure 2.6: Sequence Diagram — UC-06: Checkout and Payment

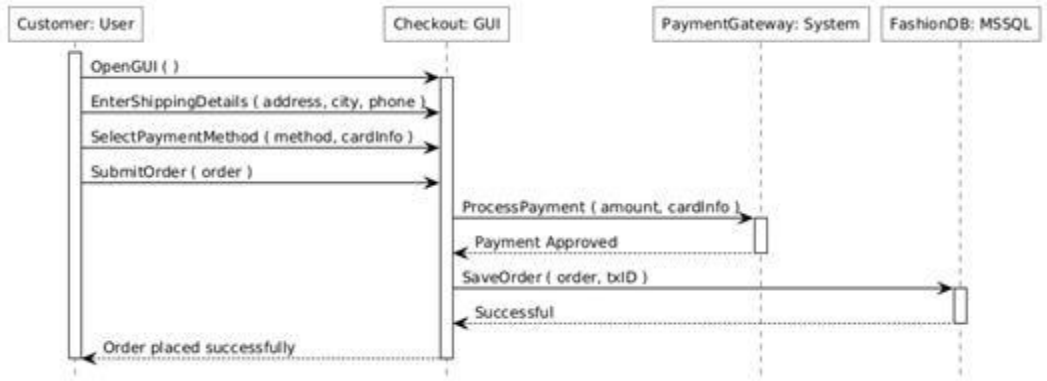


Table 2.6: Action Steps & Presentation Flow for UC-06

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Customer: User	Checkout: GUI	**OpenGUI ( )**	Normal	Customer initiates checkout from shopping cart.
2	Customer: User	Checkout: GUI	**EnterShippingDetails (address, city, phone)**	Normal	Customer enters physical shipping destination and contact.
3	Customer: User	Checkout: GUI	**SelectPaymentMethod (method, cardInfo)**	Normal	Customer chooses payment method

					and inputs billing details.
4	Customer: User	Checkout: GUI	**SubmitOrder (order)**	Normal	Customer clicks 'Place Order' button.
5	Checkout: GUI	PaymentGateway: System	**ProcessPayment (amount, cardInfo)**	Normal	GUI initiates payment charge via payment gateway.
6	PaymentGateway: System	Checkout: GUI	**Payment Approved**	Normal	Payment gateway authorizes transaction and returns approval token.
7	Checkout: GUI	FashionDB: MSSQL	**SaveOrder (order, txID)**	Normal	GUI saves confirmed order, order items, and payment status in database.
8	FashionDB: MSSQL	Checkout: GUI	**Successful**	Normal	Database confirms order record persistence and stock deduction.
9	Checkout: GUI	Customer: User	**Order placed successfully**	Normal	GUI displays order confirmation number and receipt details.

Alternative Branches & Exceptions:

- Cash on Delivery (COD): Customer selects COD. Payment gateway charge step is bypassed and order is saved with 'Pending Payment' status.
- Payment Declined: If bank declines charge, PaymentGateway returns 'Declined', and Checkout: GUI displays 'Payment failed. Please check card details'.

2.7 UC-07: Track Order Status

Description: Customer enters their Order ID or navigates to order history to query shipment tracking milestones (Pending, Confirmed, Shipped, Out for Delivery, Delivered) and view invoice information from the database.

Actor	Customer
Trigger	Customer submits Order ID or clicks on an order in their order history.
Preconditions	• Customer possesses an existing Order ID or is logged in with active orders in FashionDB: MSSQL.

Postconditions	• Real-time order progress timeline and delivery status are displayed.

Figure 2.7: Sequence Diagram — UC-07: Track Order Status

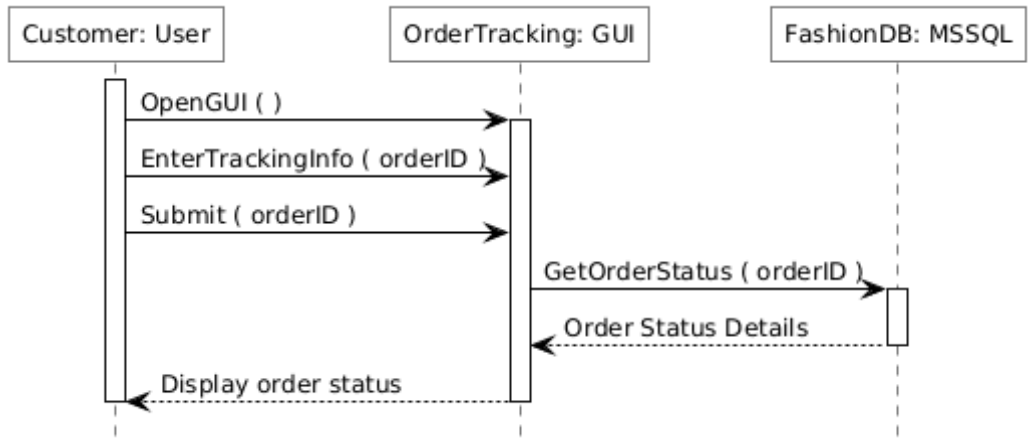


Table 2.7: Action Steps & Presentation Flow for UC-07

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Customer: User	OrderTracking: GUI	**OpenGUI ( )**	Normal	Customer opens the 'Track Order' page.
2	Customer: User	OrderTracking: GUI	**EnterTrackingInfo (orderID)**	Normal	Customer types in their unique order number.
3	Customer: User	OrderTracking: GUI	**Submit (orderID)**	Normal	Customer clicks the track button.
4	OrderTracking: GUI	FashionDB: MSSQL	**GetOrderStatus (orderID)**	Normal	GUI queries current fulfillment status from database.
5	FashionDB: MSSQL	OrderTracking: GUI	**Order Status Details**	Normal	Database returns courier info, shipment status, and timestamps.
6	OrderTracking: GUI	Customer: User	**Display order status**	Normal	GUI displays visual delivery progress stepper and item summary.

Alternative Branches & Exceptions:

- Download Invoice: Customer clicks 'Download PDF Invoice'. GUI generates and downloads tax invoice document.
- Order Not Found: If Order ID is invalid, FashionDB returns null, and OrderTracking: GUI displays 'Order ID not found. Please check and retry'.

2.8 UC-08: Feedback and Returns

Description: Customer selects a delivered order to submit product ratings/written reviews or initiates a return/refund request by providing reasons and photo evidence, which is stored in the database.

Actor	Customer
Trigger	Customer clicks 'Write Review' or 'Request Return' on a delivered order item.
Preconditions	• Target order has status 'Delivered' in FashionDB: MSSQL. • Return request is within allowable return policy window (e.g. 7 days).
Postconditions	• Product review or return request record is registered in FashionDB: MSSQL.

Figure 2.8: Sequence Diagram — UC-08: Feedback and Returns

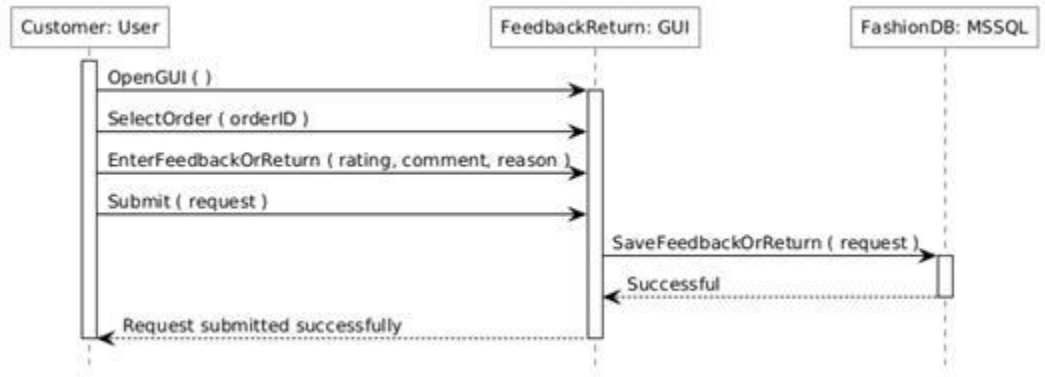


Table 2.8: Action Steps & Presentation Flow for UC-08

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Customer: User	FeedbackReturn: GUI	**OpenGUI ( )**	Normal	Customer opens the feedback and returns page.
2	Customer: User	FeedbackReturn: GUI	**SelectOrder (orderID)**	Normal	Customer chooses the delivered order item.
3	Customer: User	FeedbackReturn: GUI	**EnterFeedbackOrReturn (rating, comment, reason)**	Normal	Customer provides star rating, comments, or return reason.
4	Customer: User	FeedbackReturn: GUI	**Submit (request)**	Normal	Customer clicks submit button.

5	FeedbackReturn: GUI	FashionDB: MSSQL	**SaveFeedbackOrReturn (request)**	Normal	GUI inserts review or return ticket into database.
6	FashionDB: MSSQL	FeedbackReturn: GUI	**Successful**	Normal	Database confirms persistence of request record.
7	FeedbackReturn: GUI	Customer: User	**Request submitted successfully**	Normal	GUI displays acknowledgment message and return ticket reference.

Alternative Branches & Exceptions:

- Image Upload for Damaged Item: Customer attaches photo evidence of defective merchandise.
- Expired Return Window: If delivered date exceeds return window, GUI flags notice 'Return window expired; feedback accepted only'.

2.9 UC-09: Manage Products

Description: Store Administrator manages the fashion catalog by adding new products, editing existing apparel descriptions, prices, categories, and adjusting stock quantities in the database.

Actor	Admin
Trigger	Admin opens the Product Management panel and inputs product modifications.
Preconditions	• Admin is logged in with administrative privileges. • Admin accesses the Admin Product Management console.
Postconditions	• Product and category changes are saved in FashionDB: MSSQL.

Figure 2.9: Sequence Diagram — UC-09: Manage Products

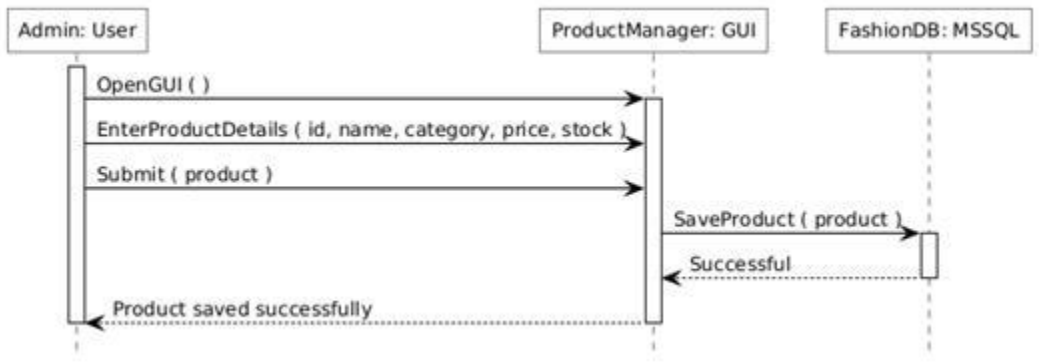


Table 2.9: Action Steps & Presentation Flow for UC-09

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Admin: User	ProductManager: GUI	**OpenGUI ( )**	Normal	Admin navigates to 'Manage Products' tab in admin portal.
2	Admin: User	ProductManager: GUI	**EnterProductDetails (id, name, category, price, stock)**	Normal	Admin inputs product code, title, category, price, and stock count.
3	Admin: User	ProductManager: GUI	**Submit (product)**	Normal	Admin clicks 'Save Product' button.
4	ProductManager: GUI	FashionDB: MSSQL	**SaveProduct (product)**	Normal	GUI submits product entity to database for insertion or update.
5	FashionDB: MSSQL	ProductManager: GUI	**Successful**	Normal	Database confirms successful product write operation.
6	ProductManager: GUI	Admin: User	**Product saved successfully**	Normal	GUI displays confirmation banner and refreshes product catalog list.

Alternative Branches & Exceptions:

- Delete / Archive Product: Admin clicks 'Delete' on an item. GUI marks product as discontinued in FashionDB: MSSQL.
- Duplicate SKU / Invalid Input: If SKU exists or price is invalid, FashionDB returns error and GUI displays 'Duplicate SKU or invalid parameters'.

2.10 UC-10: Manage Orders

Description: Store Administrator reviews incoming customer orders, inspects order items, and updates the fulfillment status (Confirmed, Processing, Shipped, Delivered, Cancelled) in the database.

Actor	Admin
Trigger	Admin opens Order Management console and selects an order to update.
Preconditions	• Admin is authenticated and on the Orders Management console. • Customer orders exist in FashionDB: MSSQL.
Postconditions	• Order status is updated in FashionDB: MSSQL and tracking info is updated.

Figure 2.10: Sequence Diagram — UC-10: Manage Orders

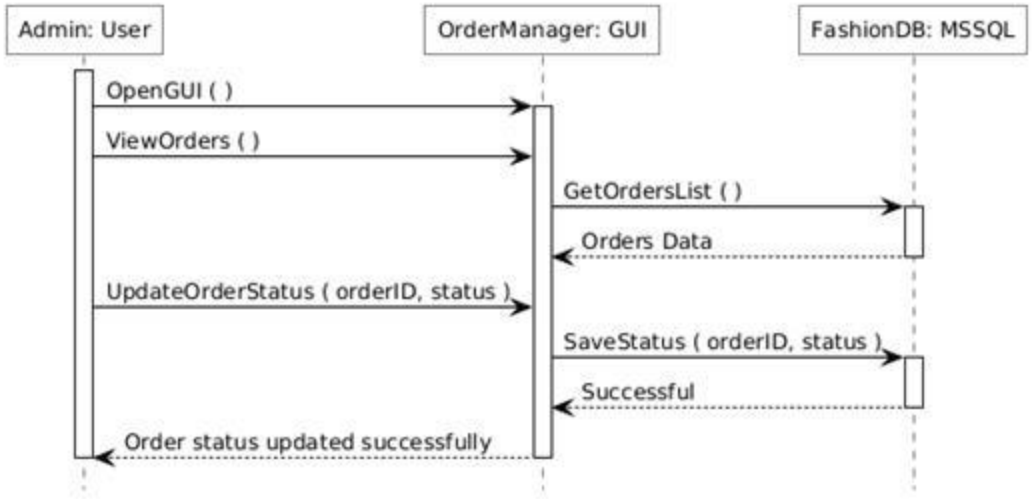


Table 2.10: Action Steps & Presentation Flow for UC-10

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Admin: User	OrderManager: GUI	**OpenGUI ( )**	Normal	Admin accesses the Order Management interface.
2	Admin: User	OrderManager: GUI	**ViewOrders ( )**	Normal	Admin clicks to load recent customer orders.
3	OrderManager: GUI	FashionDB: MSSQL	**GetOrdersList ( )**	Normal	GUI requests active orders from database.
4	FashionDB: MSSQL	OrderManager: GUI	**Orders Data**	Normal	Database returns order summaries and customer details.
5	Admin: User	OrderManager: GUI	**UpdateOrderStatus (orderID, status)**	Normal	Admin selects new status (e.g. Shipped) and inputs tracking number.
6	OrderManager: GUI	FashionDB: MSSQL	**SaveStatus (orderID, status)**	Normal	GUI transmits updated status to database.
7	FashionDB: MSSQL	OrderManager: GUI	**Successful**	Normal	Database persists updated status and timestamp.
8	OrderManager: GUI	Admin: User	**Order status updated successfully**	Normal	GUI displays success notification and updates order table.

Alternative Branches & Exceptions:

- Filter Orders by Status: Admin filters orders list by 'Pending', 'Shipped', or 'Delivered'.
- Order Cancellation Error: If customer already received goods, GUI prevents status rollback to 'Pending'.

2.11 UC-11: Manage Payments

Description: Store Administrator audits transaction logs, checks payment reconciliation records, and processes electronic refunds through the payment gateway for approved returns.

Actor	Admin
Trigger	Admin reviews payment log or approves a customer refund request.
Preconditions	• Admin is authenticated with financial clearance. • Payment transaction exists in FashionDB: MSSQL.
Postconditions	• Refund transaction is processed by PaymentGateway and status updated to 'Refunded' in database.

Figure 2.11: Sequence Diagram — UC-11: Manage Payments

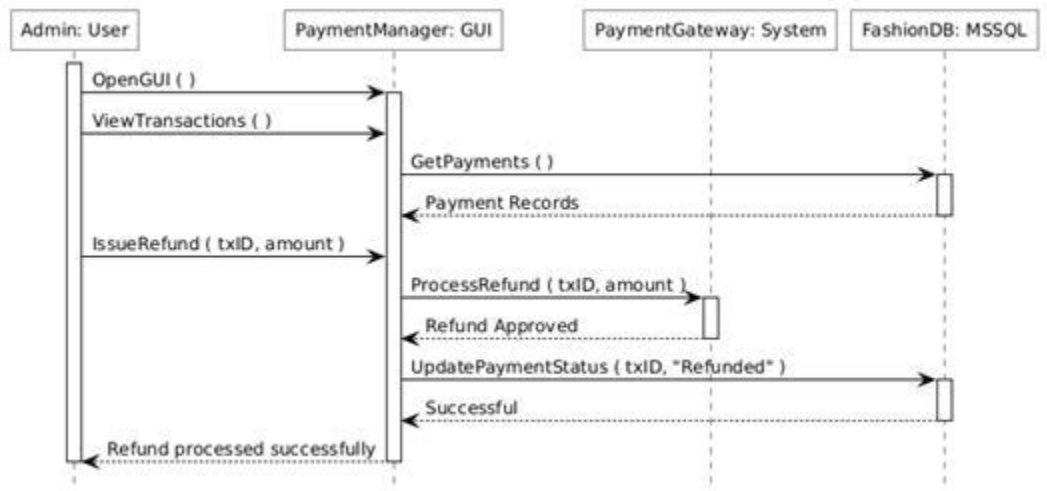


Table 2.11: Action Steps & Presentation Flow for UC-11

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Admin: User	PaymentManager: GUI	**OpenGUI ( )**	Normal	Admin navigates to Payment Management screen.

2	Admin: User	PaymentManager: GUI	**ViewTransactions ( )**	Normal	Admin clicks to inspect financial transactions.
3	PaymentManager: GUI	FashionDB: MSSQL	**GetPayments ( )**	Normal	GUI queries transaction history from database.
4	FashionDB: MSSQL	PaymentManager: GUI	**Payment Records**	Normal	Database returns payment records, gateways, and statuses.
5	Admin: User	PaymentManager: GUI	**IssueRefund (txID, amount)**	Normal	Admin selects transaction and clicks 'Process Refund'.
6	PaymentManager: GUI	PaymentGateway: System	**ProcessRefund (txID, amount)**	Normal	GUI contacts external payment gateway to issue electronic refund.
7	PaymentGateway: System	PaymentManager: GUI	**Refund Approved**	Normal	Payment gateway approves refund and returns refund reference.
8	PaymentManager: GUI	FashionDB: MSSQL	**UpdatePaymentStatus (txID, "Refunded")**	Normal	GUI updates payment record to 'Refunded' in database.
9	FashionDB: MSSQL	PaymentManager: GUI	**Successful**	Normal	Database confirms update of financial record.
10	PaymentManager: GUI	Admin: User	**Refund processed successfully**	Normal	GUI displays refund completion message and updated balance.

Alternative Branches & Exceptions:

- Export Financial Ledger: Admin downloads CSV/Excel statement of monthly settled transactions.
- Refund Declined: If payment gateway rejects refund due to expired card, GUI notifies Admin to issue manual store credit.

2.12 UC-12: Manage Customers

Description: Store Administrator inspects customer accounts, views registered contact information and order history, and manages customer account statuses (Active, Suspended, Flagged) in the database.

Actor	Admin

Trigger	Admin opens Customer Management module.
Preconditions	• Admin is logged in to the admin console. • Customer accounts exist in FashionDB: MSSQL.
Postconditions	• Customer account status or profile modifications are saved in database.

Figure 2.12: Sequence Diagram — UC-12: Manage Customers

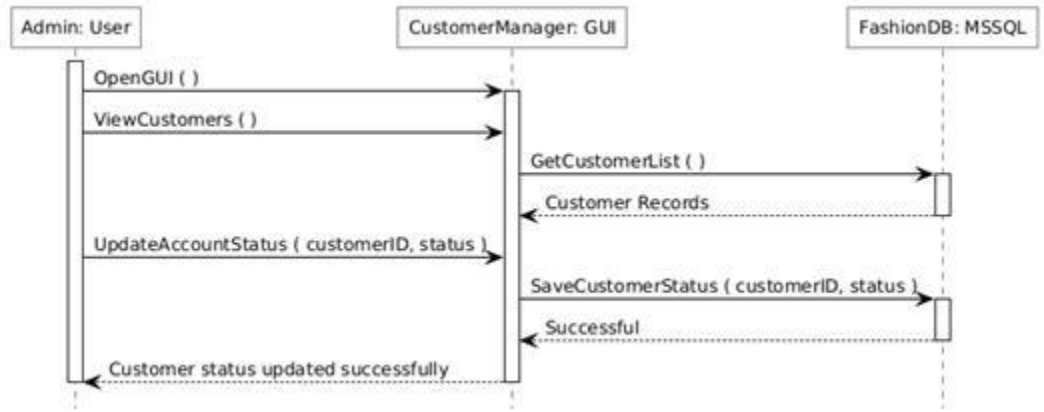


Table 2.12: Action Steps & Presentation Flow for UC-12

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Admin: User	CustomerManager: GUI	**OpenGUI ( )**	Normal	Admin navigates to Customer Management tab.
2	Admin: User	CustomerManager: GUI	**ViewCustomers ( )**	Normal	Admin requests customer list.
3	CustomerManager: GUI	FashionDB: MSSQL	**GetCustomerList ( )**	Normal	GUI queries registered customer records from database.
4	FashionDB: MSSQL	CustomerManager: GUI	**Customer Records**	Normal	Database returns customer data (names, emails, join dates, statuses).
5	Admin: User	CustomerManager: GUI	**UpdateAccountStatus (customerID, status)**	Normal	Admin selects account and updates status (e.g. Active / Suspended).

6	CustomerManager: GUI	FashionDB: MSSQL	**SaveCustomerStatus (customerID, status)**	Normal	GUI sends updated account status to database.
7	FashionDB: MSSQL	CustomerManager: GUI	**Successful**	Normal	Database confirms customer status update.
8	CustomerManager: GUI	Admin: User	**Customer status updated successfully**	Normal	GUI displays confirmation notice and refreshes customer table.

Alternative Branches & Exceptions:

- Search Customer by Email/Phone: Admin enters customer email to isolate a single user record.
- Account Not Found: If customer ID does not exist, GUI flags error 'Customer record not found'.

2.13 UC-13: Manage Promotions

Description: Store Administrator configures promotional campaigns, discount coupon codes, percentage discounts, minimum order thresholds, and expiry dates in the database.

Actor	Admin
Trigger	Admin enters promotional parameters and clicks 'Save Promotion'.
Preconditions	• Admin is authenticated with promotion management rights. • Admin accesses the Promotions Management console.
Postconditions	• Promotional campaign or coupon code is persisted in FashionDB: MSSQL.

Figure 2.13: Sequence Diagram — UC-13: Manage Promotions

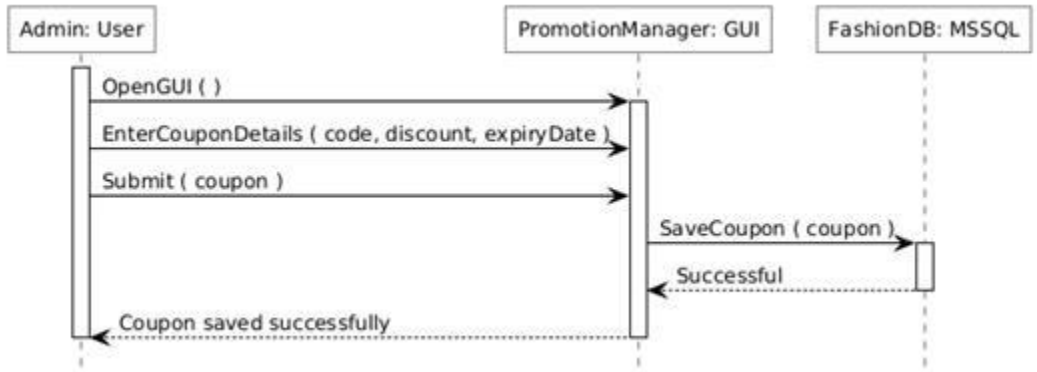


Table 2.13: Action Steps & Presentation Flow for UC-13

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Admin: User	PromotionManager: GUI	**OpenGUI ( )**	Normal	Admin opens the Promotions & Discounts management module.
2	Admin: User	PromotionManager: GUI	**EnterCouponDetails (code, discount, expiryDate)**	Normal	Admin inputs coupon code, discount percentage, and expiry date.
3	Admin: User	PromotionManager: GUI	**Submit (coupon)**	Normal	Admin clicks 'Create Promotion' button.
4	PromotionManager: GUI	FashionDB: MSSQL	**SaveCoupon (coupon)**	Normal	GUI transmits coupon entity to database.
5	FashionDB: MSSQL	PromotionManager: GUI	**Successful**	Normal	Database confirms insertion of promotion record.
6	PromotionManager: GUI	Admin: User	**Coupon saved successfully**	Normal	GUI displays success notification and updates promotions table.

Alternative Branches & Exceptions:

- Deactivate Promotion: Admin toggles off an active promotion code prior to its scheduled expiry.
- Duplicate Promotion Code: If coupon code already exists, FashionDB returns error and GUI prompts to choose a unique code.

2.14 UC-14: Generate Reports

Description: Store Administrator selects analytics criteria (date range, product category, sales performance, inventory turnover) to generate structured business reports from the database.

Actor	Admin
Trigger	Admin selects report criteria and clicks 'Generate Report'.
Preconditions	• Admin is logged in to the admin reporting portal. • Historical sales and inventory records exist in FashionDB: MSSQL.
Postconditions	• Analytical report is compiled, displayed, and ready for export.

Figure 2.14: Sequence Diagram — UC-14: Generate Reports

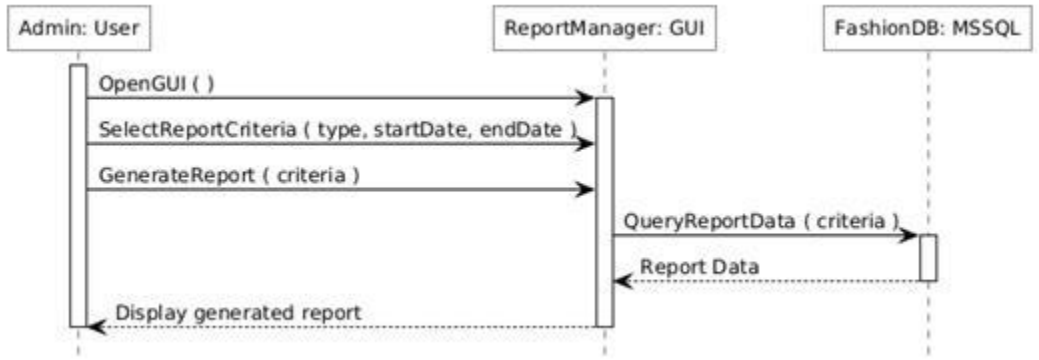


Table 2.14: Action Steps & Presentation Flow for UC-14

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Admin: User	ReportManager: GUI	**OpenGUI ( )**	Normal	Admin navigates to the 'Reports & Analytics' dashboard.
2	Admin: User	ReportManager: GUI	**SelectReportCriteria (type, startDate, endDate)**	Normal	Admin selects report type (Sales, Inventory, Orders) and date span.
3	Admin: User	ReportManager: GUI	**GenerateReport (criteria)**	Normal	Admin clicks 'Generate Report' button.
4	ReportManager: GUI	FashionDB: MSSQL	**QueryReportData (criteria)**	Normal	GUI queries aggregated analytics data from database.
5	FashionDB: MSSQL	ReportManager: GUI	**Report Data**	Normal	Database returns revenue metrics, order quantities, and product stats.
6	ReportManager: GUI	Admin: User	**Display generated report**	Normal	GUI renders charts, data tables, and download options.

Alternative Branches & Exceptions:

- Export Report: Admin clicks 'Export as PDF/CSV' to save report file locally.
- No Data for Date Range: If no transactions occurred within selected span, GUI displays 'No data recorded for this time range'.

2.15 UC-15: Manage Profile and Shipping Address

Description: Customer views their account profile, updates personal information (name, phone), and adds, edits, or deletes saved shipping addresses in the database.

Actor	Customer
Trigger	Customer modifies profile details or enters a new shipping address and clicks 'Save'.
Preconditions	• Customer is logged in to the online fashion store. • Customer accesses the 'My Account / Profile' section.
Postconditions	• Updated personal information and shipping addresses are stored in FashionDB: MSSQL.

Figure 2.15: Sequence Diagram — UC-15: Manage Profile and Shipping Address

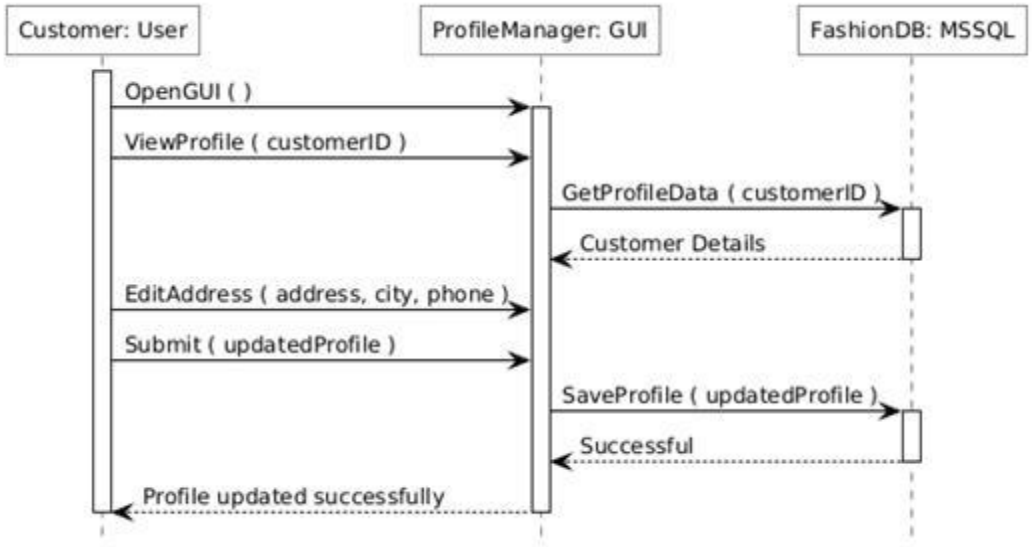


Table 2.15: Action Steps & Presentation Flow for UC-15

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Customer: User	ProfileManager: GUI	**OpenGUI ( )**	Normal	Customer opens the 'Profile & Addresses' page.
2	Customer: User	ProfileManager: GUI	**ViewProfile (customerID)**	Normal	Customer views personal details and address card list.
3	ProfileManager: GUI	FashionDB: MSSQL	**GetProfileData (customerID)**	Normal	GUI fetches current profile and address

					entries from database.
4	FashionDB: MSSQL	ProfileManager: GUI	**Customer Details**	Normal	Database returns customer entity and address book records.
5	Customer: User	ProfileManager: GUI	**EditAddress (address, city, phone)**	Normal	Customer modifies address details or adds a new address.
6	Customer: User	ProfileManager: GUI	**Submit (updatedProfile)**	Normal	Customer clicks 'Save Profile Changes'.
7	ProfileManager: GUI	FashionDB: MSSQL	**SaveProfile (updatedProfile)**	Normal	GUI transmits updated entity to database.
8	FashionDB: MSSQL	ProfileManager: GUI	**Successful**	Normal	Database confirms persistence of updated customer information.
9	ProfileManager: GUI	Customer: User	**Profile updated successfully**	Normal	GUI displays confirmation banner and refreshes address view.

Alternative Branches & Exceptions:

- Set Default Shipping Address: Customer toggles primary address for 1-click checkout.
- Validation Failure: If postal code or phone number format is invalid, GUI displays field validation notice.

3. Requirements Traceability Matrix (RTM)

The following matrix maps every Functional Requirement from Lab 01 SRS to the corresponding presentation-level Use Case and Sequence Diagram.

Requirement ID	Functional Requirement Statement (SRS Lab 01)	Mapped Use Case & Sequence Diagram
FR-01	Customers can register and log in to their account.	**UC-01 (Register), UC-02 (Login)**
FR-02	Customers can browse products by category (men, women, kids, etc.).	**UC-03 (Browse & Search)**
FR-03	Customers can search and filter products (by size, color, price).	**UC-03 (Browse & Search)**

FR-04	Customers can view product details (images, price, description, sizes).	**UC-04 (View Product Info)**
FR-05	Customers can add/remove products to/from the shopping cart.	**UC-05 (Manage Cart)**
FR-06	Customers can place an order and choose a payment method.	**UC-06 (Checkout & Payment)**
FR-07	Customers can make payment via multiple options (card, COD, wallet).	**UC-06 (Checkout & Payment)**
FR-08	Customers can track the status of their order (pending, shipped, delivered).	**UC-07 (Track Order Status)**
FR-09	Customers can view order history and download invoices/receipts.	**UC-07 (Track Order Status)**
FR-10	Customers can request a return/refund for a delivered item.	**UC-08 (Feedback & Returns)**
FR-11	Customers can edit their profile and shipping address.	**UC-15 (Manage Profile & Shipping Address)**
FR-12	Customers can leave reviews and ratings on purchased products.	**UC-08 (Feedback & Returns)**
FR-13	Admin can add, edit, or remove products and categories.	**UC-09 (Manage Products)**
FR-14	Admin can manage inventory and stock levels.	**UC-09 (Manage Products)**
FR-15	Admin can view and manage customer orders (confirm, ship, cancel).	**UC-10 (Manage Orders)**
FR-16	Admin can update order status for tracking purposes.	**UC-10 (Manage Orders)**
FR-17	Admin can manage payments and process refunds.	**UC-11 (Manage Payments)**
FR-18	Admin can manage customer accounts.	**UC-12 (Manage Customers)**
FR-19	Admin can generate sales and inventory reports.	**UC-14 (Generate Reports)**
FR-20	Admin can manage discounts/promotional offers.	**UC-13 (Manage Promotions)**

SYSTEM DESIGN & PRESENTATION SPECIFICATION

ONLINE FASHION STORE

UML Sequence Diagrams & Use Case Interaction Manual

A complete, presentation-ready UML Sequence Diagram specification for the Online Fashion Store system. Each diagram maps directly to the project's Use Case specifications, illustrating the dynamic interactions between actors (Customer / Admin), User Interface (UI), System Controller, and Database. Specifically structured for academic project documentation, slide presentations, and viva defense.

Project Title	Online Fashion Store (Ecommerce-Fashion)
System Domain	Single-Vendor Web-Based E-Commerce Platform
Diagram Modeling Level	Academic Presentation Level (Actor: User -> Screen: GUI -> FashionDB: MSSQL)
Document Version	2.0 (Presentation Edition)
Requirements Coverage	100% Coverage of Lab 01 Functional Requirements (FR-01 to FR-20)
Documentation Scope	All Customer & Admin Use Cases (UC-01 through UC-15)

1. Sequence Diagram Presentation Pattern

In software project presentations, final viva defenses, and academic evaluation panels, sequence diagrams must clearly communicate the chronological flow of actions for each use case. Rather than cluttering diagrams with low-level implementation details or distributed

network frames, the presentation-level pattern models the essential interaction flow between the user, the graphical user interface, and the database.

1.1 Academic Presentation Architecture (Actor -> GUI -> Database)

Lifeline Element	UML Role	Presentation & Viva Explanation
Actor: User (Customer / Admin)	Human Initiator	The actor initiating the use case via client web browser or administrative management console.
<Module>: GUI (Boundary)	Presentation Boundary	The specific user interface screen or modal form (e.g., Register: GUI, Cart: GUI, ProductManager: GUI) capturing input and displaying responses.
FashionDB: MSSQL (Data Store)	Persistent Entity Store	The relational database management system maintaining products, customer accounts, orders, and configuration records.
PaymentGateway: System (External)	Financial Service	External banking/payment gateway service processing electronic card charges and automated refunds (UC-06 & UC-11).

2. Use Case Sequence Diagrams (Complete Catalog)

The following catalog details all Use Cases for the Online Fashion Store. Each section contains the use case description, actor, preconditions, postconditions, an embedded presentation sequence diagram visual, a step-by-step trace table, and alternative and exception branch analysis.

2.1 UC-01: Register

Description: Customer registers for a new store account by opening the registration form, providing personal information (name, email, password, contact), validating email uniqueness against the database, and saving the customer profile.

Actor	Customer
Trigger	Customer navigates to the registration screen and submits their registration details.
Preconditions	• Customer has accessed the online fashion store web portal. • Customer has a valid, unique email address.

Postconditions	• New customer record is saved in FashionDB: MSSQL. • Customer is registered and redirected to login/homepage.

Figure 2.1: Sequence Diagram — UC-01: Register

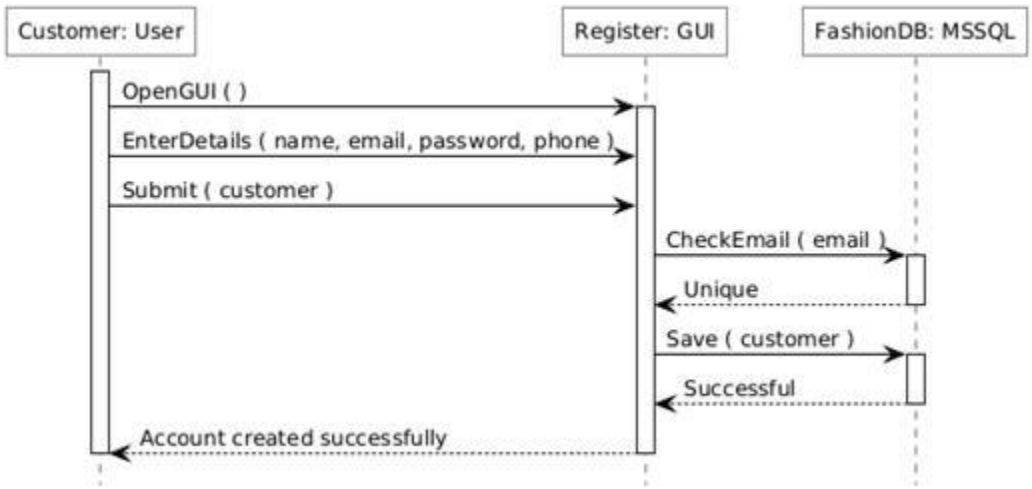


Table 2.1: Action Steps & Presentation Flow for UC-01

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Customer: User	Register: GUI	**OpenGUI ( )**	Normal	Customer clicks register link on store navigation header.
2	Customer: User	Register: GUI	**EnterDetails (name, email, password, phone)**	Normal	Customer fills out the account registration input fields.
3	Customer: User	Register: GUI	**Submit (customer)**	Normal	Customer clicks the submit button to register.
4	Register: GUI	FashionDB: MSSQL	**CheckEmail (email)**	Normal	GUI queries the database to verify email uniqueness.
5	FashionDB: MSSQL	Register: GUI	**Unique**	Normal	Database confirms the provided email is not previously registered.
6	Register: GUI	FashionDB: MSSQL	**Save (customer)**	Normal	GUI transmits customer entity to be inserted into database.

7	FashionDB: MSSQL	Register: GUI	**Successful**	Normal	Database returns success status for customer insertion.
8	Register: GUI	Customer: User	**Account created successfully**	Normal	GUI displays confirmation message and redirects customer.

Alternative Branches & Exceptions:

- Social Media Registration: Customer selects 'Sign up with Google'. GUI authenticates external identity and saves profile directly in FashionDB: MSSQL.
- Duplicate Email: If email already exists, FashionDB returns 'Email Exists', and Register: GUI displays 'Email already registered. Please login'.

2.2 UC-02: Login

Description: Customer logs in to their existing account by providing registered email and password credentials, which are verified against the database before granting account access.

Actor	Customer
Trigger	Customer enters credentials and clicks 'Login'.
Preconditions	• Customer possesses an existing registered account in FashionDB: MSSQL. • Customer has accessed the login screen.
Postconditions	• Customer session is authenticated and authorized. • Customer receives access to account features and personalized dashboard.

Figure 2.2: Sequence Diagram — UC-02: Login

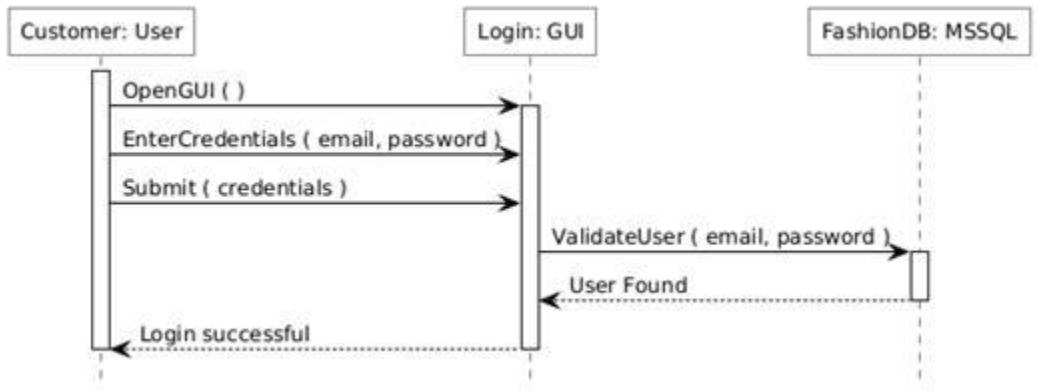


Table 2.2: Action Steps & Presentation Flow for UC-02

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Customer: User	Login: GUI	**OpenGUI ( )**	Normal	Customer clicks login button on store navigation bar.
2	Customer: User	Login: GUI	**EnterCredentials (email, password)**	Normal	Customer types email and password into login fields.
3	Customer: User	Login: GUI	**Submit (credentials)**	Normal	Customer clicks the login submit button.
4	Login: GUI	FashionDB: MSSQL	**ValidateUser (email, password)**	Normal	GUI queries database to verify user credentials.
5	FashionDB: MSSQL	Login: GUI	**User Found**	Normal	Database verifies credentials and returns matching user profile.
6	Login: GUI	Customer: User	**Login successful**	Normal	GUI creates user session and redirects customer to homepage.

Alternative Branches & Exceptions:

- Password Reset Flow: Customer clicks 'Forgot Password'. Login GUI prompts for email and triggers verification link.
- Invalid Credentials: If email or password is invalid, FashionDB returns 'Invalid', and Login: GUI displays 'Invalid credentials. Please try again'.

2.3 UC-03: Browse and Search

Description: Customer explores fashion products by selecting categories (men, women, kids) and entering search queries with faceted filters (price, size, color) to retrieve matching products from the database.

Actor	Customer
Trigger	Customer selects a catalog category or inputs search keywords.
Preconditions	• Fashion store catalog is populated with active apparel items in FashionDB: MSSQL.
Postconditions	• Filtered fashion product catalog is presented on the customer screen.

Figure 2.3: Sequence Diagram — UC-03: Browse and Search

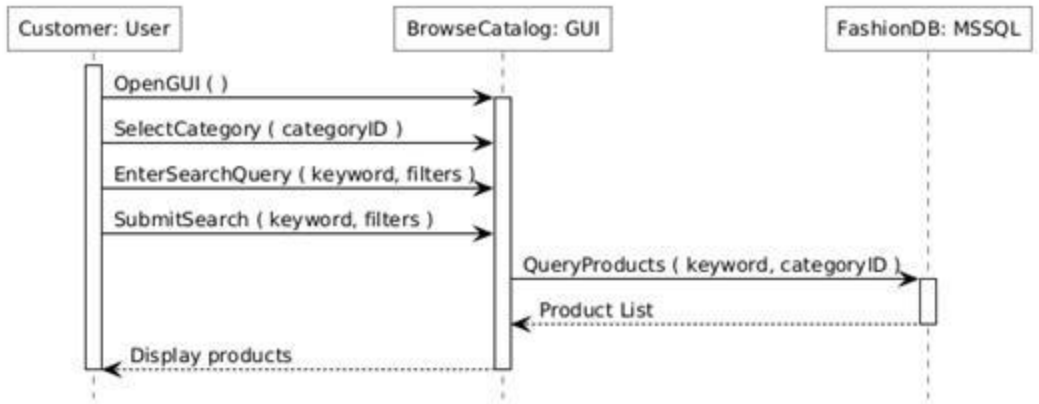


Table 2.3: Action Steps & Presentation Flow for UC-03

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Customer: User	BrowseCatalog: GUI	**OpenGUI ( )**	Normal	Customer accesses the store storefront / catalog page.
2	Customer: User	BrowseCatalog: GUI	**SelectCategory (categoryID)**	Normal	Customer selects apparel category (e.g., Men, Women, Kids).
3	Customer: User	BrowseCatalog: GUI	**EnterSearchQuery (keyword, filters)**	Normal	Customer specifies search term, size, color, or price range.
4	Customer: User	BrowseCatalog: GUI	**SubmitSearch (keyword, filters)**	Normal	Customer triggers the search filtering operation.
5	BrowseCatalog: GUI	FashionDB: MSSQL	**QueryProducts (keyword, categoryID)**	Normal	GUI queries product catalog table in FashionDB.
6	FashionDB: MSSQL	BrowseCatalog: GUI	**Product List**	Normal	Database returns list of matched product records.
7	BrowseCatalog: GUI	Customer: User	**Display products**	Normal	GUI displays product grid matching customer criteria.

Alternative Branches & Exceptions:

- Sorting by Price/Popularity: Customer selects sort criteria (e.g. Price: Low to High). GUI re-queries database with sorting parameters.
- No Products Found: If query matches zero products, FashionDB returns empty set, and GUI displays 'No products found matching your filters'.

2.4 UC-04: View Product Info

Description: Customer clicks on a specific product card to view comprehensive item specifications, high-definition image galleries, available sizes, stock status, pricing, and authentic customer reviews retrieved from the database.

Actor	Customer
Trigger	Customer clicks on a product card or thumbnail.
Preconditions	• Customer is browsing the catalog or search results. • Selected product exists in FashionDB: MSSQL.
Postconditions	• Detailed product view and customer ratings/reviews are displayed.

Figure 2.4: Sequence Diagram — UC-04: View Product Info

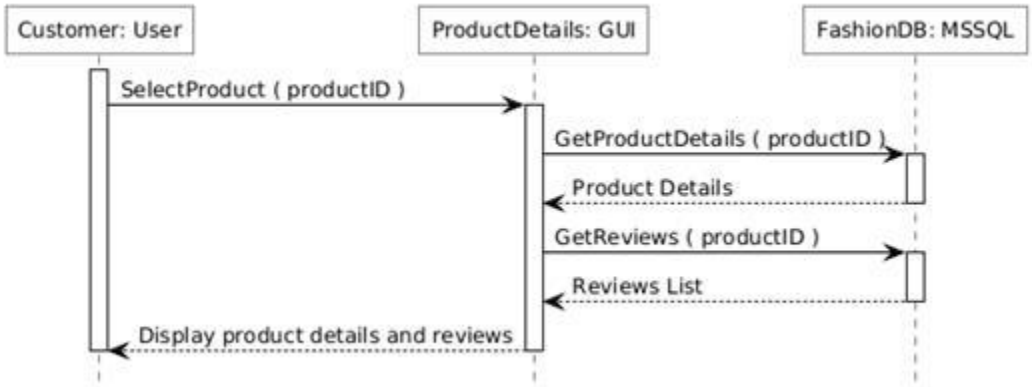


Table 2.4: Action Steps & Presentation Flow for UC-04

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Customer: User	ProductDetails: GUI	**SelectProduct (productID)**	Normal	Customer clicks on an apparel item card in the catalog.
2	ProductDetails: GUI	FashionDB: MSSQL	**GetProductDetails (productID)**	Normal	GUI fetches detailed product attributes from database.
3	FashionDB: MSSQL	ProductDetails: GUI	**Product Details**	Normal	Database returns images, description, sizes, and price.

4	ProductDetails: GUI	FashionDB: MSSQL	**GetReviews (productID)**	Normal	GUI queries ratings and customer feedback for the product.
5	FashionDB: MSSQL	ProductDetails: GUI	**Reviews List**	Normal	Database returns list of approved customer reviews.
6	ProductDetails: GUI	Customer: User	**Display product details and reviews**	Normal	GUI presents detailed product view with image gallery and reviews.

Alternative Branches & Exceptions:

- Zoom & Gallery View: Customer clicks to inspect high-resolution fabric texture images.
- Product Inactive/Removed: If item is discontinued, FashionDB returns null, and GUI displays 'This product is no longer available'.

2.5 UC-05: Manage Cart

Description: Customer adds fashion items to their shopping cart, verifies item inventory availability in the database, adjusts item quantities, or removes items prior to proceeding to checkout.

Actor	Customer
Trigger	Customer clicks 'Add to Cart', adjusts item quantity, or opens the cart view.
Preconditions	• Customer is viewing a product page or cart drawer. • Items added have valid size and quantity selections.
Postconditions	• Shopping cart items and running total are updated and saved in FashionDB: MSSQL.

Figure 2.5: Sequence Diagram — UC-05: Manage Cart

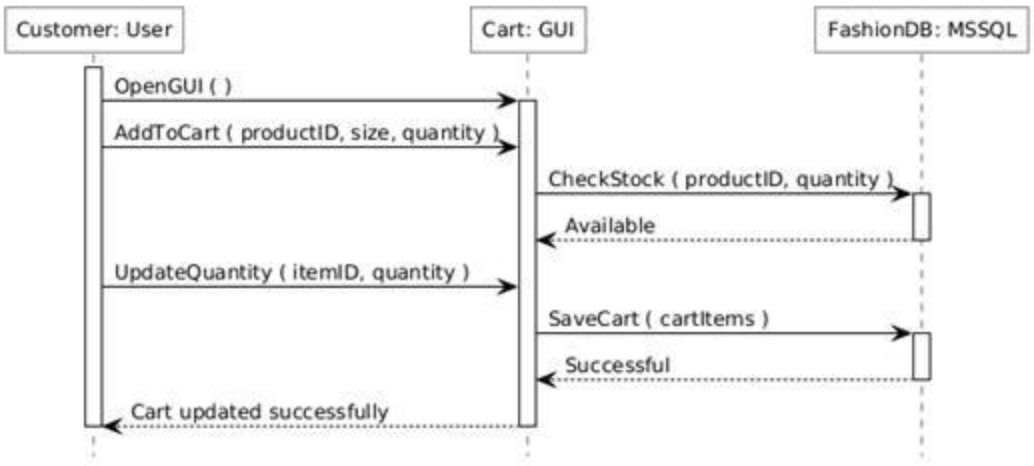


Table 2.5: Action Steps & Presentation Flow for UC-05

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Customer: User	Cart: GUI	**OpenGUI ( )**	Normal	Customer opens the shopping cart panel.
2	Customer: User	Cart: GUI	**AddToCart (productID, size, quantity)**	Normal	Customer adds apparel item with chosen size and quantity.
3	Cart: GUI	FashionDB: MSSQL	**CheckStock (productID, quantity)**	Normal	GUI queries stock level in database.
4	FashionDB: MSSQL	Cart: GUI	**Available**	Normal	Database confirms required inventory is in stock.
5	Customer: User	Cart: GUI	**UpdateQuantity (itemID, quantity)**	Normal	Customer adjusts item count or removes item.
6	Cart: GUI	FashionDB: MSSQL	**SaveCart (cartItems)**	Normal	GUI persists updated cart contents in database.
7	FashionDB: MSSQL	Cart: GUI	**Successful**	Normal	Database confirms cart changes are saved.
8	Cart: GUI	Customer: User	**Cart updated successfully**	Normal	GUI updates subtotal and displays updated cart list.

Alternative Branches & Exceptions:

- Remove Item from Cart: Customer clicks 'Remove'. GUI updates cart session and refreshes subtotal.
- Out of Stock: If requested quantity exceeds warehouse stock, FashionDB returns 'Insufficient Stock', and Cart: GUI displays warning.

2.6 UC-06: Checkout and Payment

Description: Customer provides shipping address, selects payment method (Credit/Debit Card, Cash on Delivery, Mobile Wallet), authorizes transaction through the payment gateway, and persists the confirmed order in the database.

Actor	Customer
Trigger	Customer clicks 'Proceed to Checkout' and submits payment details.
Preconditions	• Customer has one or more valid items in their shopping cart. • Customer provides complete delivery and payment information.
Postconditions	• Order is confirmed with unique Order ID and payment transaction record in FashionDB: MSSQL. • Inventory stock is decremented accordingly.

Figure 2.6: Sequence Diagram — UC-06: Checkout and Payment

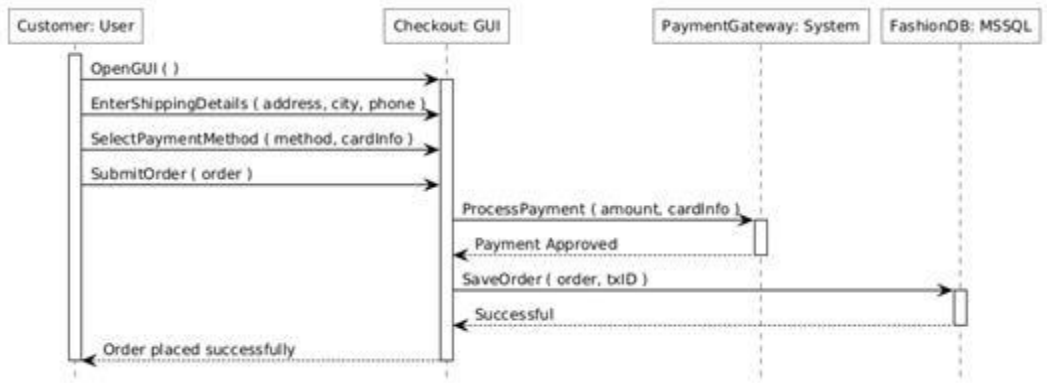


Table 2.6: Action Steps & Presentation Flow for UC-06

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Customer: User	Checkout: GUI	**OpenGUI ( )**	Normal	Customer initiates checkout from shopping cart.
2	Customer: User	Checkout: GUI	**EnterShippingDetails (address, city, phone)**	Normal	Customer enters physical shipping destination and contact.
3	Customer: User	Checkout: GUI	**SelectPaymentMethod (method, cardInfo)**	Normal	Customer chooses payment method

					and inputs billing details.
4	Customer: User	Checkout: GUI	**SubmitOrder (order)**	Normal	Customer clicks 'Place Order' button.
5	Checkout: GUI	PaymentGateway: System	**ProcessPayment (amount, cardInfo)**	Normal	GUI initiates payment charge via payment gateway.
6	PaymentGateway: System	Checkout: GUI	**Payment Approved**	Normal	Payment gateway authorizes transaction and returns approval token.
7	Checkout: GUI	FashionDB: MSSQL	**SaveOrder (order, txID)**	Normal	GUI saves confirmed order, order items, and payment status in database.
8	FashionDB: MSSQL	Checkout: GUI	**Successful**	Normal	Database confirms order record persistence and stock deduction.
9	Checkout: GUI	Customer: User	**Order placed successfully**	Normal	GUI displays order confirmation number and receipt details.

Alternative Branches & Exceptions:

- Cash on Delivery (COD): Customer selects COD. Payment gateway charge step is bypassed and order is saved with 'Pending Payment' status.
- Payment Declined: If bank declines charge, PaymentGateway returns 'Declined', and Checkout: GUI displays 'Payment failed. Please check card details'.

2.7 UC-07: Track Order Status

Description: Customer enters their Order ID or navigates to order history to query shipment tracking milestones (Pending, Confirmed, Shipped, Out for Delivery, Delivered) and view invoice information from the database.

Actor	Customer
Trigger	Customer submits Order ID or clicks on an order in their order history.
Preconditions	• Customer possesses an existing Order ID or is logged in with active orders in FashionDB: MSSQL.

Postconditions	• Real-time order progress timeline and delivery status are displayed.

Figure 2.7: Sequence Diagram — UC-07: Track Order Status

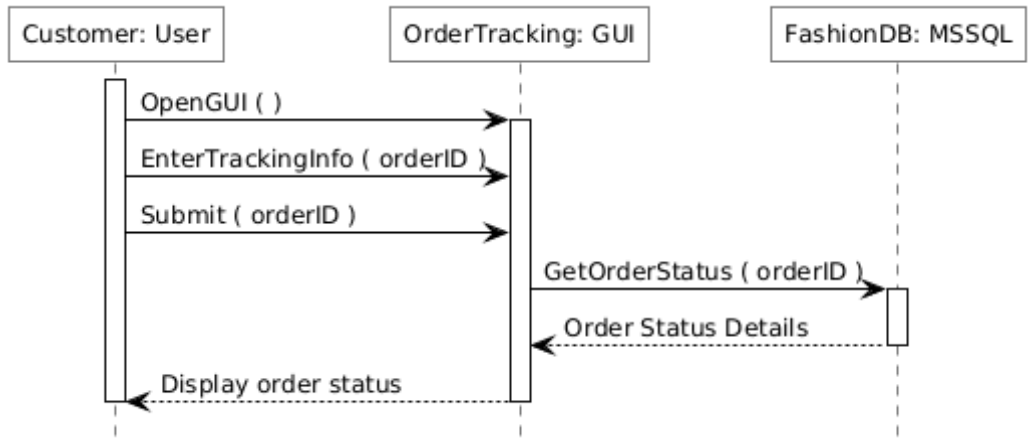


Table 2.7: Action Steps & Presentation Flow for UC-07

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Customer: User	OrderTracking: GUI	**OpenGUI ( )**	Normal	Customer opens the 'Track Order' page.
2	Customer: User	OrderTracking: GUI	**EnterTrackingInfo (orderID)**	Normal	Customer types in their unique order number.
3	Customer: User	OrderTracking: GUI	**Submit (orderID)**	Normal	Customer clicks the track button.
4	OrderTracking: GUI	FashionDB: MSSQL	**GetOrderStatus (orderID)**	Normal	GUI queries current fulfillment status from database.
5	FashionDB: MSSQL	OrderTracking: GUI	**Order Status Details**	Normal	Database returns courier info, shipment status, and timestamps.
6	OrderTracking: GUI	Customer: User	**Display order status**	Normal	GUI displays visual delivery progress stepper and item summary.

Alternative Branches & Exceptions:

- Download Invoice: Customer clicks 'Download PDF Invoice'. GUI generates and downloads tax invoice document.
- Order Not Found: If Order ID is invalid, FashionDB returns null, and OrderTracking: GUI displays 'Order ID not found. Please check and retry'.

2.8 UC-08: Feedback and Returns

Description: Customer selects a delivered order to submit product ratings/written reviews or initiates a return/refund request by providing reasons and photo evidence, which is stored in the database.

Actor	Customer
Trigger	Customer clicks 'Write Review' or 'Request Return' on a delivered order item.
Preconditions	• Target order has status 'Delivered' in FashionDB: MSSQL. • Return request is within allowable return policy window (e.g. 7 days).
Postconditions	• Product review or return request record is registered in FashionDB: MSSQL.

Figure 2.8: Sequence Diagram — UC-08: Feedback and Returns

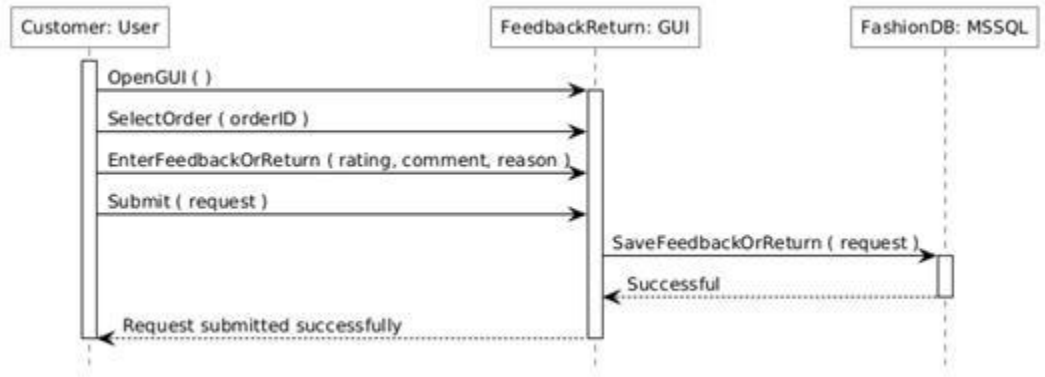


Table 2.8: Action Steps & Presentation Flow for UC-08

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Customer: User	FeedbackReturn: GUI	**OpenGUI ( )**	Normal	Customer opens the feedback and returns page.
2	Customer: User	FeedbackReturn: GUI	**SelectOrder (orderID)**	Normal	Customer chooses the delivered order item.
3	Customer: User	FeedbackReturn: GUI	**EnterFeedbackOrReturn (rating, comment, reason)**	Normal	Customer provides star rating, comments, or return reason.
4	Customer: User	FeedbackReturn: GUI	**Submit (request)**	Normal	Customer clicks submit button.

5	FeedbackReturn: GUI	FashionDB: MSSQL	**SaveFeedbackOrReturn (request)**	Normal	GUI inserts review or return ticket into database.
6	FashionDB: MSSQL	FeedbackReturn: GUI	**Successful**	Normal	Database confirms persistence of request record.
7	FeedbackReturn: GUI	Customer: User	**Request submitted successfully**	Normal	GUI displays acknowledgment message and return ticket reference.

Alternative Branches & Exceptions:

- Image Upload for Damaged Item: Customer attaches photo evidence of defective merchandise.
- Expired Return Window: If delivered date exceeds return window, GUI flags notice 'Return window expired; feedback accepted only'.

2.9 UC-09: Manage Products

Description: Store Administrator manages the fashion catalog by adding new products, editing existing apparel descriptions, prices, categories, and adjusting stock quantities in the database.

Actor	Admin
Trigger	Admin opens the Product Management panel and inputs product modifications.
Preconditions	• Admin is logged in with administrative privileges. • Admin accesses the Admin Product Management console.
Postconditions	• Product and category changes are saved in FashionDB: MSSQL.

Figure 2.9: Sequence Diagram — UC-09: Manage Products

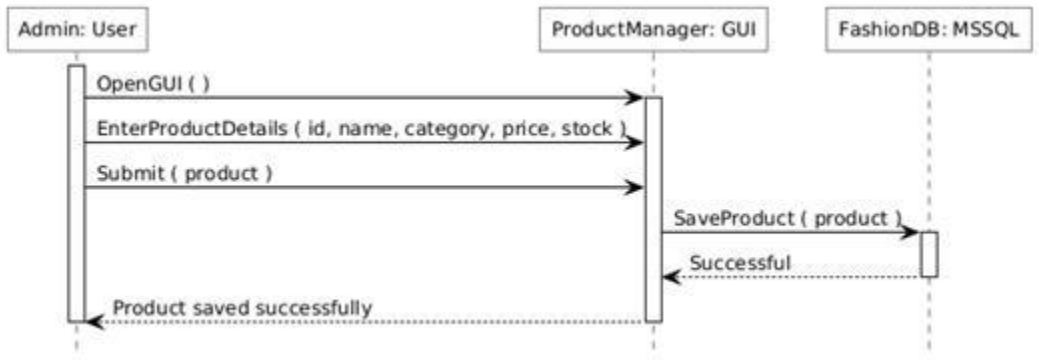


Table 2.9: Action Steps & Presentation Flow for UC-09

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Admin: User	ProductManager: GUI	**OpenGUI ( )**	Normal	Admin navigates to 'Manage Products' tab in admin portal.
2	Admin: User	ProductManager: GUI	**EnterProductDetails (id, name, category, price, stock)**	Normal	Admin inputs product code, title, category, price, and stock count.
3	Admin: User	ProductManager: GUI	**Submit (product)**	Normal	Admin clicks 'Save Product' button.
4	ProductManager: GUI	FashionDB: MSSQL	**SaveProduct (product)**	Normal	GUI submits product entity to database for insertion or update.
5	FashionDB: MSSQL	ProductManager: GUI	**Successful**	Normal	Database confirms successful product write operation.
6	ProductManager: GUI	Admin: User	**Product saved successfully**	Normal	GUI displays confirmation banner and refreshes product catalog list.

Alternative Branches & Exceptions:

- Delete / Archive Product: Admin clicks 'Delete' on an item. GUI marks product as discontinued in FashionDB: MSSQL.
- Duplicate SKU / Invalid Input: If SKU exists or price is invalid, FashionDB returns error and GUI displays 'Duplicate SKU or invalid parameters'.

2.10 UC-10: Manage Orders

Description: Store Administrator reviews incoming customer orders, inspects order items, and updates the fulfillment status (Confirmed, Processing, Shipped, Delivered, Cancelled) in the database.

Actor	Admin
Trigger	Admin opens Order Management console and selects an order to update.
Preconditions	• Admin is authenticated and on the Orders Management console. • Customer orders exist in FashionDB: MSSQL.
Postconditions	• Order status is updated in FashionDB: MSSQL and tracking info is updated.

Figure 2.10: Sequence Diagram — UC-10: Manage Orders

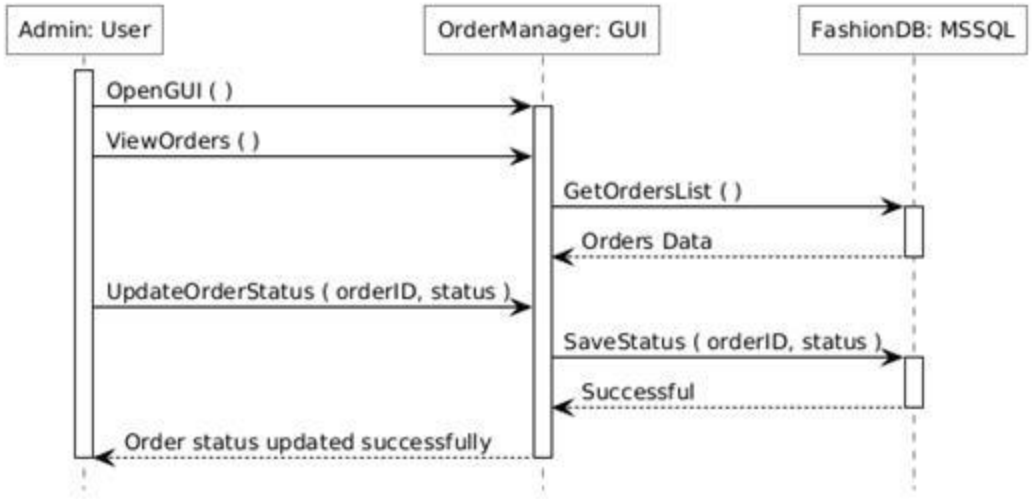


Table 2.10: Action Steps & Presentation Flow for UC-10

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Admin: User	OrderManager: GUI	**OpenGUI ( )**	Normal	Admin accesses the Order Management interface.
2	Admin: User	OrderManager: GUI	**ViewOrders ( )**	Normal	Admin clicks to load recent customer orders.
3	OrderManager: GUI	FashionDB: MSSQL	**GetOrdersList ( )**	Normal	GUI requests active orders from database.
4	FashionDB: MSSQL	OrderManager: GUI	**Orders Data**	Normal	Database returns order summaries and customer details.
5	Admin: User	OrderManager: GUI	**UpdateOrderStatus (orderID, status)**	Normal	Admin selects new status (e.g. Shipped) and inputs tracking number.
6	OrderManager: GUI	FashionDB: MSSQL	**SaveStatus (orderID, status)**	Normal	GUI transmits updated status to database.
7	FashionDB: MSSQL	OrderManager: GUI	**Successful**	Normal	Database persists updated status and timestamp.
8	OrderManager: GUI	Admin: User	**Order status updated successfully**	Normal	GUI displays success notification and updates order table.

Alternative Branches & Exceptions:

- Filter Orders by Status: Admin filters orders list by 'Pending', 'Shipped', or 'Delivered'.
- Order Cancellation Error: If customer already received goods, GUI prevents status rollback to 'Pending'.

2.11 UC-11: Manage Payments

Description: Store Administrator audits transaction logs, checks payment reconciliation records, and processes electronic refunds through the payment gateway for approved returns.

Actor	Admin
Trigger	Admin reviews payment log or approves a customer refund request.
Preconditions	• Admin is authenticated with financial clearance. • Payment transaction exists in FashionDB: MSSQL.
Postconditions	• Refund transaction is processed by PaymentGateway and status updated to 'Refunded' in database.

Figure 2.11: Sequence Diagram — UC-11: Manage Payments

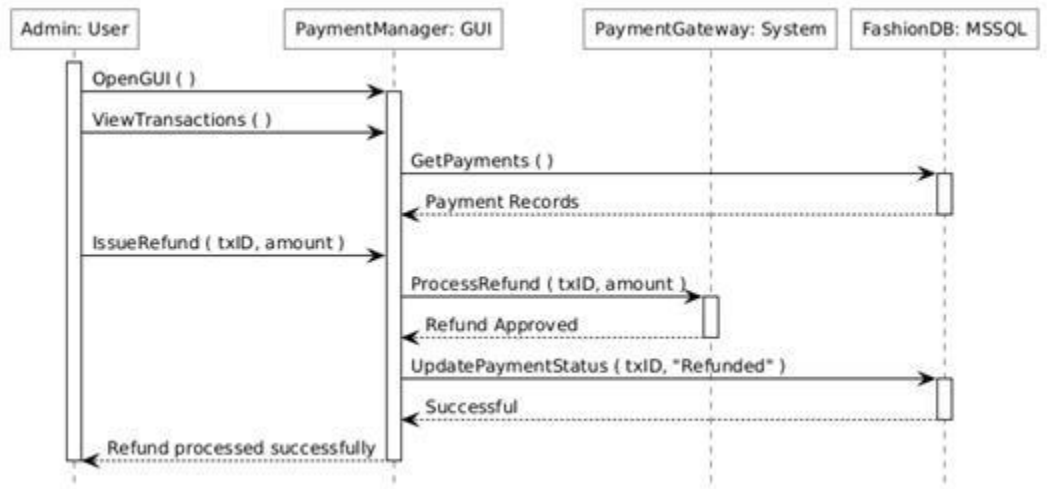


Table 2.11: Action Steps & Presentation Flow for UC-11

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Admin: User	PaymentManager: GUI	**OpenGUI ( )**	Normal	Admin navigates to Payment Management screen.

2	Admin: User	PaymentManager: GUI	**ViewTransactions ( )**	Normal	Admin clicks to inspect financial transactions.
3	PaymentManager: GUI	FashionDB: MSSQL	**GetPayments ( )**	Normal	GUI queries transaction history from database.
4	FashionDB: MSSQL	PaymentManager: GUI	**Payment Records**	Normal	Database returns payment records, gateways, and statuses.
5	Admin: User	PaymentManager: GUI	**IssueRefund (txID, amount)**	Normal	Admin selects transaction and clicks 'Process Refund'.
6	PaymentManager: GUI	PaymentGateway: System	**ProcessRefund (txID, amount)**	Normal	GUI contacts external payment gateway to issue electronic refund.
7	PaymentGateway: System	PaymentManager: GUI	**Refund Approved**	Normal	Payment gateway approves refund and returns refund reference.
8	PaymentManager: GUI	FashionDB: MSSQL	**UpdatePaymentStatus (txID, "Refunded")**	Normal	GUI updates payment record to 'Refunded' in database.
9	FashionDB: MSSQL	PaymentManager: GUI	**Successful**	Normal	Database confirms update of financial record.
10	PaymentManager: GUI	Admin: User	**Refund processed successfully**	Normal	GUI displays refund completion message and updated balance.

Alternative Branches & Exceptions:

- Export Financial Ledger: Admin downloads CSV/Excel statement of monthly settled transactions.
- Refund Declined: If payment gateway rejects refund due to expired card, GUI notifies Admin to issue manual store credit.

2.12 UC-12: Manage Customers

Description: Store Administrator inspects customer accounts, views registered contact information and order history, and manages customer account statuses (Active, Suspended, Flagged) in the database.

Actor	Admin

Trigger	Admin opens Customer Management module.
Preconditions	• Admin is logged in to the admin console. • Customer accounts exist in FashionDB: MSSQL.
Postconditions	• Customer account status or profile modifications are saved in database.

Figure 2.12: Sequence Diagram — UC-12: Manage Customers

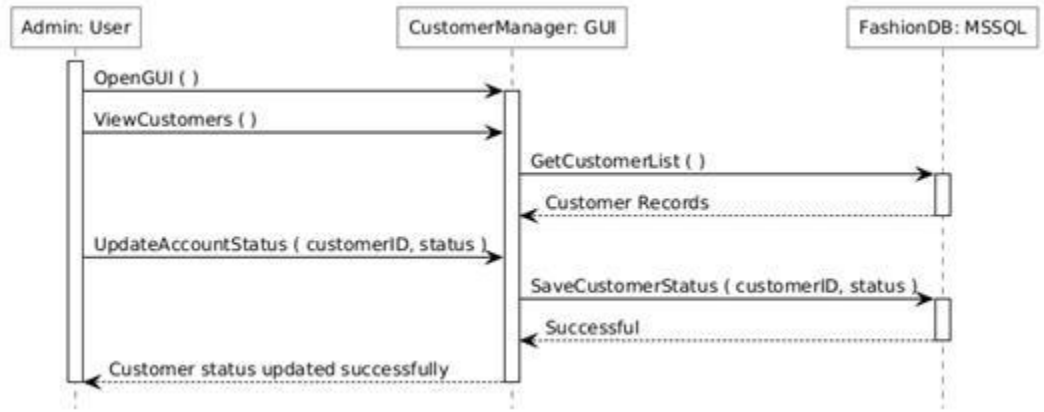


Table 2.12: Action Steps & Presentation Flow for UC-12

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Admin: User	CustomerManager: GUI	**OpenGUI ( )**	Normal	Admin navigates to Customer Management tab.
2	Admin: User	CustomerManager: GUI	**ViewCustomers ( )**	Normal	Admin requests customer list.
3	CustomerManager: GUI	FashionDB: MSSQL	**GetCustomerList ( )**	Normal	GUI queries registered customer records from database.
4	FashionDB: MSSQL	CustomerManager: GUI	**Customer Records**	Normal	Database returns customer data (names, emails, join dates, statuses).
5	Admin: User	CustomerManager: GUI	**UpdateAccountStatus (customerID, status)**	Normal	Admin selects account and updates status (e.g. Active / Suspended).

6	CustomerManager: GUI	FashionDB: MSSQL	**SaveCustomerStatus (customerID, status)**	Normal	GUI sends updated account status to database.
7	FashionDB: MSSQL	CustomerManager: GUI	**Successful**	Normal	Database confirms customer status update.
8	CustomerManager: GUI	Admin: User	**Customer status updated successfully**	Normal	GUI displays confirmation notice and refreshes customer table.

Alternative Branches & Exceptions:

- Search Customer by Email/Phone: Admin enters customer email to isolate a single user record.
- Account Not Found: If customer ID does not exist, GUI flags error 'Customer record not found'.

2.13 UC-13: Manage Promotions

Description: Store Administrator configures promotional campaigns, discount coupon codes, percentage discounts, minimum order thresholds, and expiry dates in the database.

Actor	Admin
Trigger	Admin enters promotional parameters and clicks 'Save Promotion'.
Preconditions	• Admin is authenticated with promotion management rights. • Admin accesses the Promotions Management console.
Postconditions	• Promotional campaign or coupon code is persisted in FashionDB: MSSQL.

Figure 2.13: Sequence Diagram — UC-13: Manage Promotions

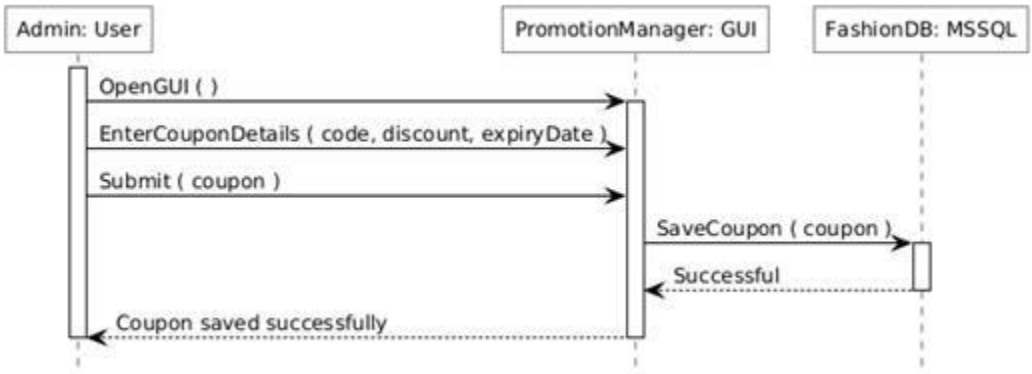


Table 2.13: Action Steps & Presentation Flow for UC-13

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Admin: User	PromotionManager: GUI	**OpenGUI ( )**	Normal	Admin opens the Promotions & Discounts management module.
2	Admin: User	PromotionManager: GUI	**EnterCouponDetails (code, discount, expiryDate)**	Normal	Admin inputs coupon code, discount percentage, and expiry date.
3	Admin: User	PromotionManager: GUI	**Submit (coupon)**	Normal	Admin clicks 'Create Promotion' button.
4	PromotionManager: GUI	FashionDB: MSSQL	**SaveCoupon (coupon)**	Normal	GUI transmits coupon entity to database.
5	FashionDB: MSSQL	PromotionManager: GUI	**Successful**	Normal	Database confirms insertion of promotion record.
6	PromotionManager: GUI	Admin: User	**Coupon saved successfully**	Normal	GUI displays success notification and updates promotions table.

Alternative Branches & Exceptions:

- Deactivate Promotion: Admin toggles off an active promotion code prior to its scheduled expiry.
- Duplicate Promotion Code: If coupon code already exists, FashionDB returns error and GUI prompts to choose a unique code.

2.14 UC-14: Generate Reports

Description: Store Administrator selects analytics criteria (date range, product category, sales performance, inventory turnover) to generate structured business reports from the database.

Actor	Admin
Trigger	Admin selects report criteria and clicks 'Generate Report'.
Preconditions	• Admin is logged in to the admin reporting portal. • Historical sales and inventory records exist in FashionDB: MSSQL.
Postconditions	• Analytical report is compiled, displayed, and ready for export.

Figure 2.14: Sequence Diagram — UC-14: Generate Reports

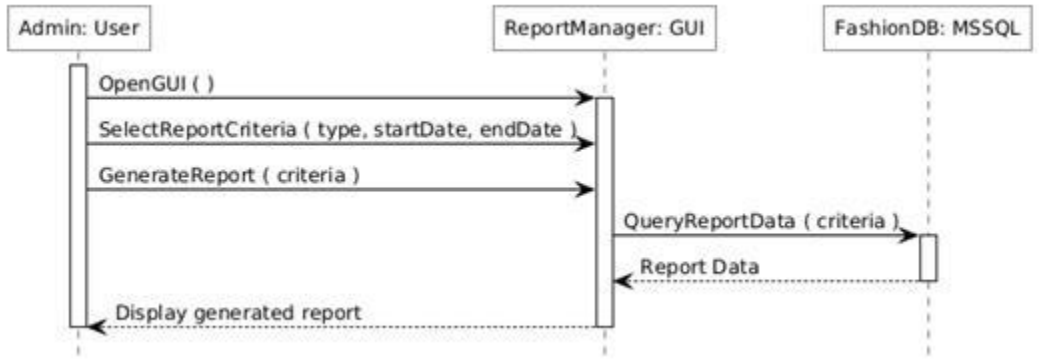


Table 2.14: Action Steps & Presentation Flow for UC-14

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Admin: User	ReportManager: GUI	**OpenGUI ( )**	Normal	Admin navigates to the 'Reports & Analytics' dashboard.
2	Admin: User	ReportManager: GUI	**SelectReportCriteria (type, startDate, endDate)**	Normal	Admin selects report type (Sales, Inventory, Orders) and date span.
3	Admin: User	ReportManager: GUI	**GenerateReport (criteria)**	Normal	Admin clicks 'Generate Report' button.
4	ReportManager: GUI	FashionDB: MSSQL	**QueryReportData (criteria)**	Normal	GUI queries aggregated analytics data from database.
5	FashionDB: MSSQL	ReportManager: GUI	**Report Data**	Normal	Database returns revenue metrics, order quantities, and product stats.
6	ReportManager: GUI	Admin: User	**Display generated report**	Normal	GUI renders charts, data tables, and download options.

Alternative Branches & Exceptions:

- Export Report: Admin clicks 'Export as PDF/CSV' to save report file locally.
- No Data for Date Range: If no transactions occurred within selected span, GUI displays 'No data recorded for this time range'.

2.15 UC-15: Manage Profile and Shipping Address

Description: Customer views their account profile, updates personal information (name, phone), and adds, edits, or deletes saved shipping addresses in the database.

Actor	Customer
Trigger	Customer modifies profile details or enters a new shipping address and clicks 'Save'.
Preconditions	• Customer is logged in to the online fashion store. • Customer accesses the 'My Account / Profile' section.
Postconditions	• Updated personal information and shipping addresses are stored in FashionDB: MSSQL.

Figure 2.15: Sequence Diagram — UC-15: Manage Profile and Shipping Address

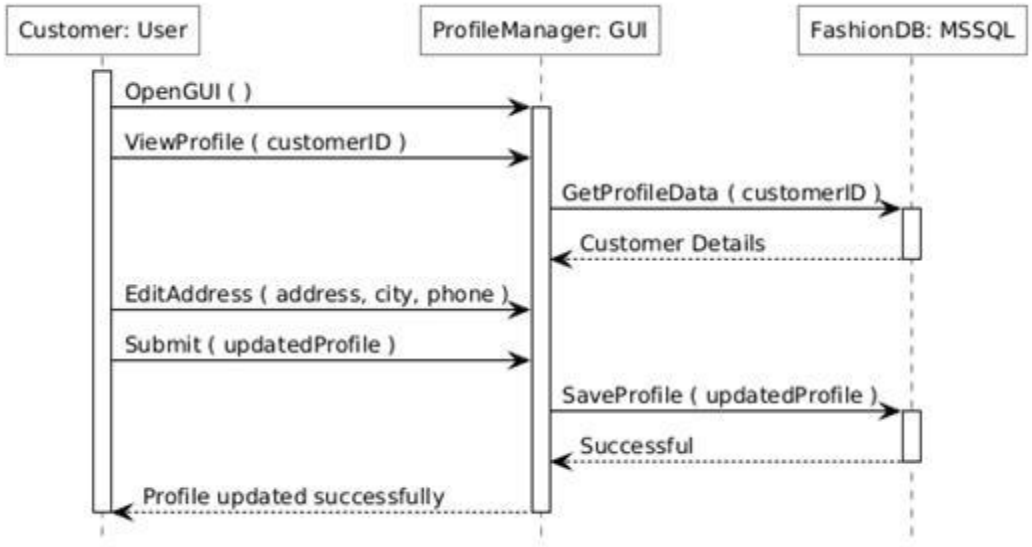


Table 2.15: Action Steps & Presentation Flow for UC-15

Step	Source	Target	Action / Message	Flow	Presentation & Viva Explanation
1	Customer: User	ProfileManager: GUI	**OpenGUI ( )**	Normal	Customer opens the 'Profile & Addresses' page.
2	Customer: User	ProfileManager: GUI	**ViewProfile (customerID)**	Normal	Customer views personal details and address card list.
3	ProfileManager: GUI	FashionDB: MSSQL	**GetProfileData (customerID)**	Normal	GUI fetches current profile and address

					entries from database.
4	FashionDB: MSSQL	ProfileManager: GUI	**Customer Details**	Normal	Database returns customer entity and address book records.
5	Customer: User	ProfileManager: GUI	**EditAddress (address, city, phone)**	Normal	Customer modifies address details or adds a new address.
6	Customer: User	ProfileManager: GUI	**Submit (updatedProfile)**	Normal	Customer clicks 'Save Profile Changes'.
7	ProfileManager: GUI	FashionDB: MSSQL	**SaveProfile (updatedProfile)**	Normal	GUI transmits updated entity to database.
8	FashionDB: MSSQL	ProfileManager: GUI	**Successful**	Normal	Database confirms persistence of updated customer information.
9	ProfileManager: GUI	Customer: User	**Profile updated successfully**	Normal	GUI displays confirmation banner and refreshes address view.

Alternative Branches & Exceptions:

- Set Default Shipping Address: Customer toggles primary address for 1-click checkout.
- Validation Failure: If postal code or phone number format is invalid, GUI displays field validation notice.

3. Requirements Traceability Matrix (RTM)

The following matrix maps every Functional Requirement from Lab 01 SRS to the corresponding presentation-level Use Case and Sequence Diagram.

Requirement ID	Functional Requirement Statement (SRS Lab 01)	Mapped Use Case & Sequence Diagram
FR-01	Customers can register and log in to their account.	**UC-01 (Register), UC-02 (Login)**
FR-02	Customers can browse products by category (men, women, kids, etc.).	**UC-03 (Browse & Search)**
FR-03	Customers can search and filter products (by size, color, price).	**UC-03 (Browse & Search)**

FR-04	Customers can view product details (images, price, description, sizes).	**UC-04 (View Product Info)**
FR-05	Customers can add/remove products to/from the shopping cart.	**UC-05 (Manage Cart)**
FR-06	Customers can place an order and choose a payment method.	**UC-06 (Checkout & Payment)**
FR-07	Customers can make payment via multiple options (card, COD, wallet).	**UC-06 (Checkout & Payment)**
FR-08	Customers can track the status of their order (pending, shipped, delivered).	**UC-07 (Track Order Status)**
FR-09	Customers can view order history and download invoices/receipts.	**UC-07 (Track Order Status)**
FR-10	Customers can request a return/refund for a delivered item.	**UC-08 (Feedback & Returns)**
FR-11	Customers can edit their profile and shipping address.	**UC-15 (Manage Profile & Shipping Address)**
FR-12	Customers can leave reviews and ratings on purchased products.	**UC-08 (Feedback & Returns)**
FR-13	Admin can add, edit, or remove products and categories.	**UC-09 (Manage Products)**
FR-14	Admin can manage inventory and stock levels.	**UC-09 (Manage Products)**
FR-15	Admin can view and manage customer orders (confirm, ship, cancel).	**UC-10 (Manage Orders)**
FR-16	Admin can update order status for tracking purposes.	**UC-10 (Manage Orders)**
FR-17	Admin can manage payments and process refunds.	**UC-11 (Manage Payments)**
FR-18	Admin can manage customer accounts.	**UC-12 (Manage Customers)**
FR-19	Admin can generate sales and inventory reports.	**UC-14 (Generate Reports)**
FR-20	Admin can manage discounts/promotional offers.	**UC-13 (Manage Promotions)**